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Geographic information — Observations, measurements and samples

*Informations géographiques — Observations, mesures et
échantillons*

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Contents

Page

Foreword	iv
1 Scope	1
2 Normative references	1
3 Terms and definitions	1
4 Content	1
5 Abstract Observation core package	1
5.1 Abstract Observation core overview.....	1
5.2 Defining tables.....	5
6 Abstract Sample core package	14
6.1 Abstract Sample core overview.....	14
6.2 Defining tables.....	19
7 Basic Observations package	24
7.1 Basic Observations overview.....	24
7.2 Defining tables.....	28
8 Basic Samples package	35
8.1 Basic Samples overview.....	35
8.2 Defining tables.....	39
9 Conceptual Observation schema package	45
9.1 Conceptual Observation schema overview.....	45
9.2 Defining tables.....	45
10 Conceptual Sample schema package	55
10.1 Conceptual Sample schema overview.....	55
10.2 Defining tables.....	55
11 Examples package	61
11.1 Examples overview.....	61
11.2 BasicFeaturesOnly package.....	63
11.2.1 BasicFeaturesOnly overview.....	63
11.3 Codelist realizations package.....	64
11.3.1 Codelist realizations overview.....	64
11.3.2 Defining tables.....	66
11.4 Collections package.....	67
11.4.1 Collections overview.....	67
11.4.2 Defining tables.....	68
11.5 ModelConsistency package.....	69
11.5.1 ModelConsistency overview.....	69
11.5.2 Defining tables.....	70
11.6 SamplingCascade package.....	71
11.6.1 SamplingCascade overview.....	71
11.6.2 Defining tables.....	73
11.7 ModelConsistency — Gender package.....	74
11.7.1 ModelConsistency — Gender overview.....	74
11.7.2 Defining tables.....	77

Foreword

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This document was prepared by Technical Committee ISO/TC 211, *Geographic information/Geomantics*.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

Geographic information — Observations, measurements and samples



1 Scope

This document provides descriptions to the ISO 19156 models.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org>

4 Content

5 Abstract Observation core package

5.1 Abstract Observation core overview

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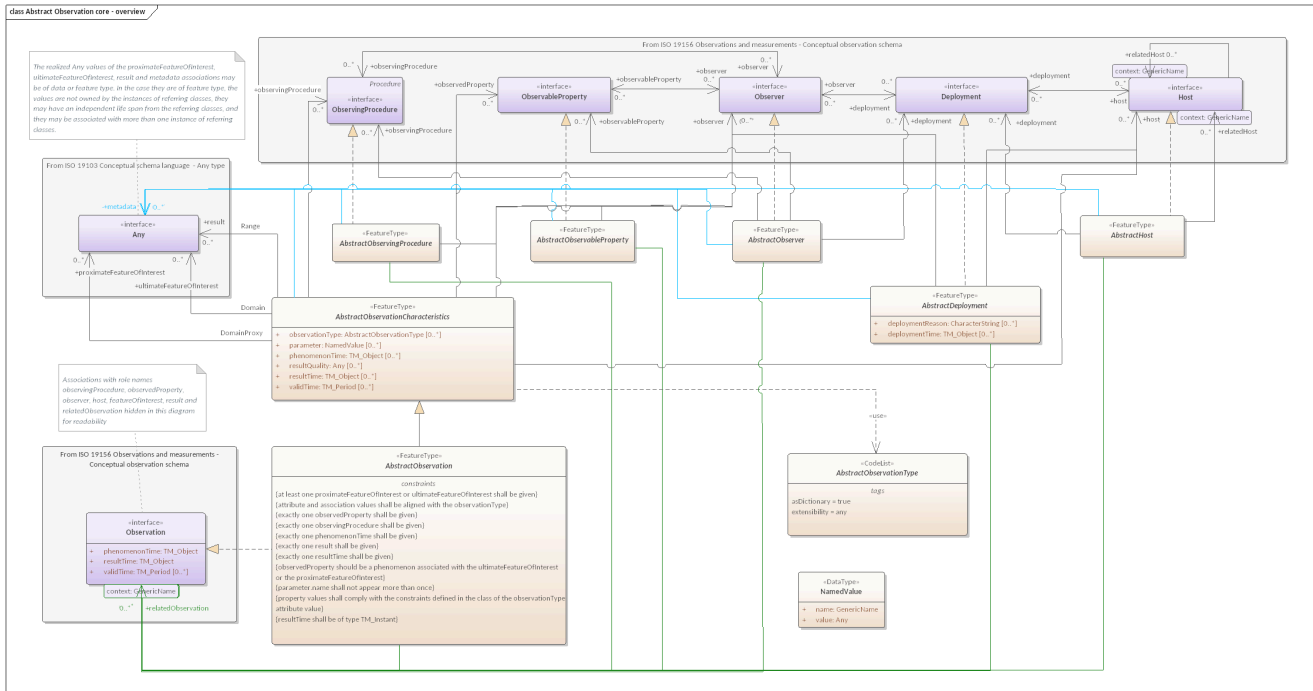


Figure 1 — Abstract Observation core - overview

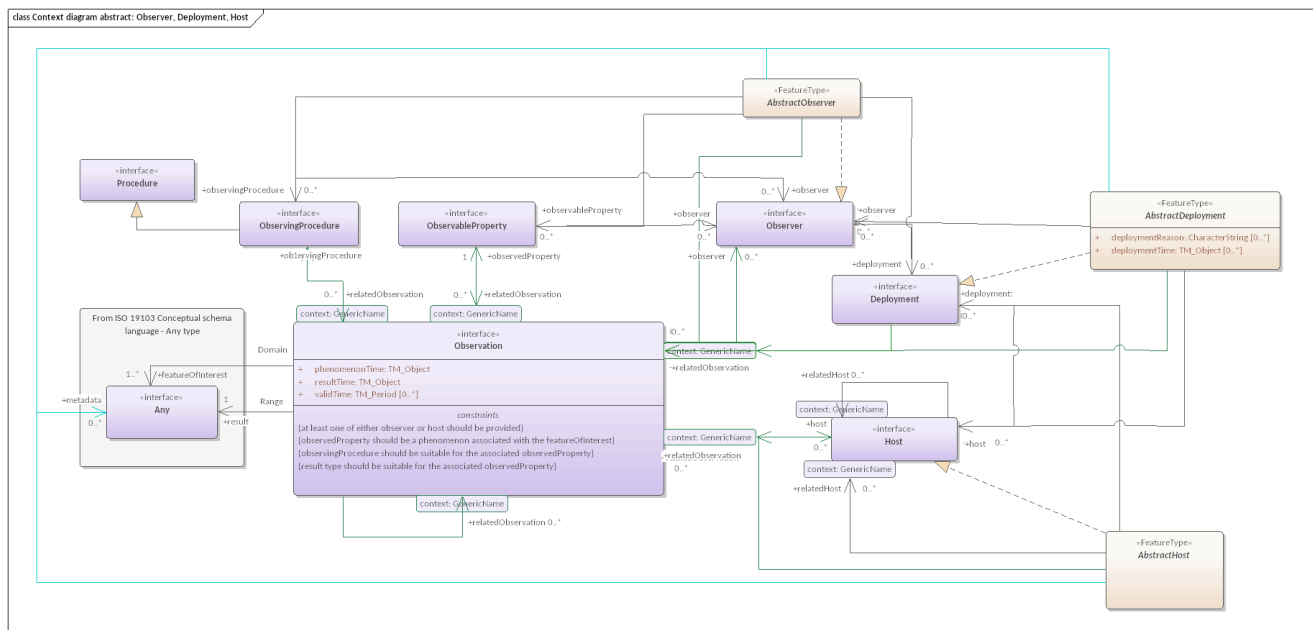


Figure 2 — Context diagram abstract: Observer, Deployment, Host

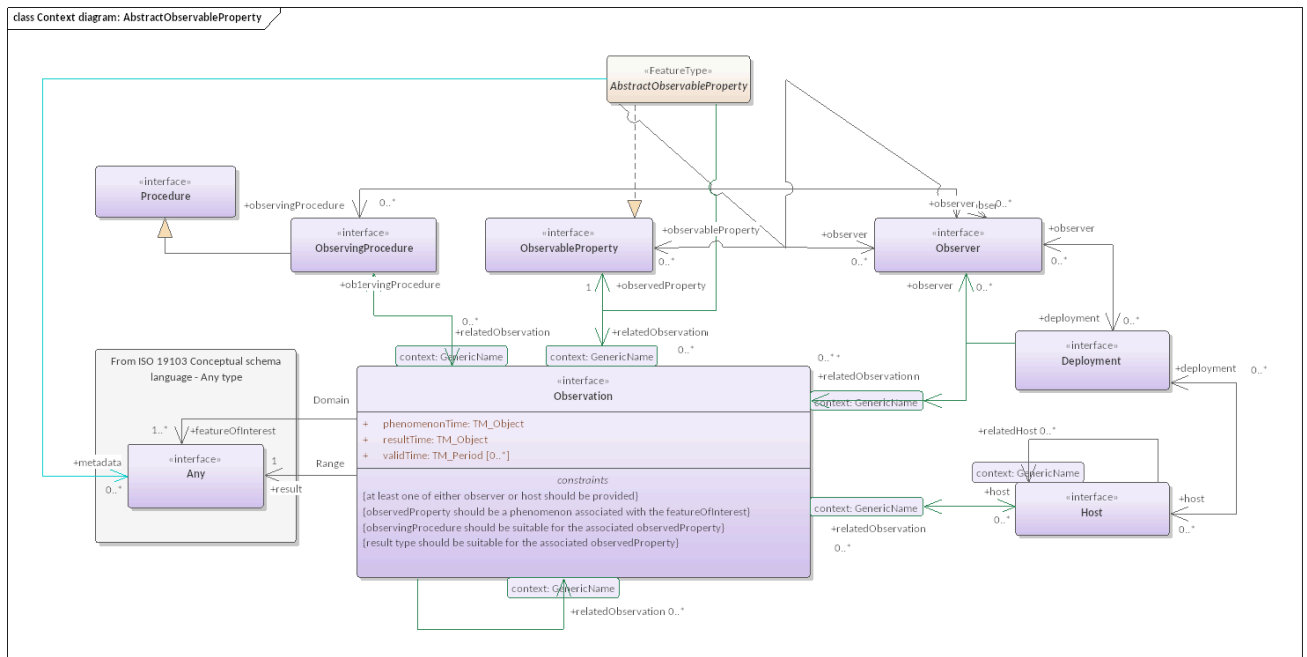


Figure 3 — Context diagram: AbstractObservableProperty

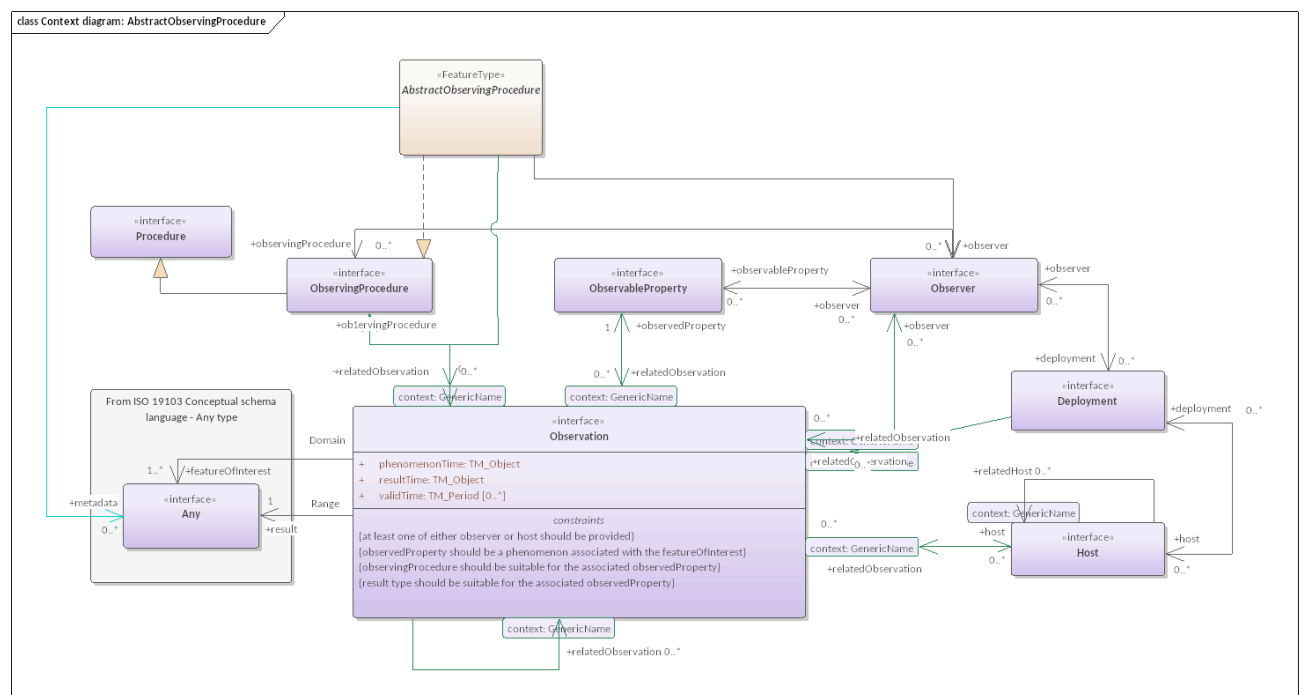


Figure 4 — Context diagram: AbstractObservingProcedure

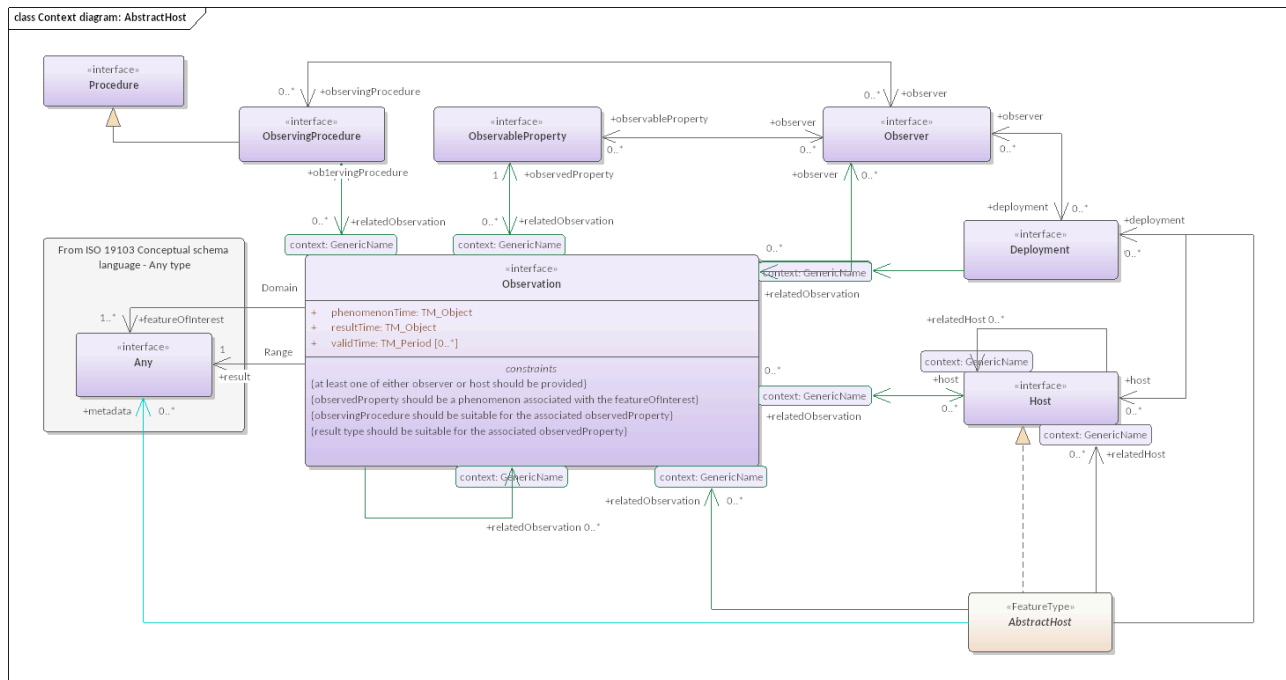


Figure 5 — Context diagram: AbstractHost

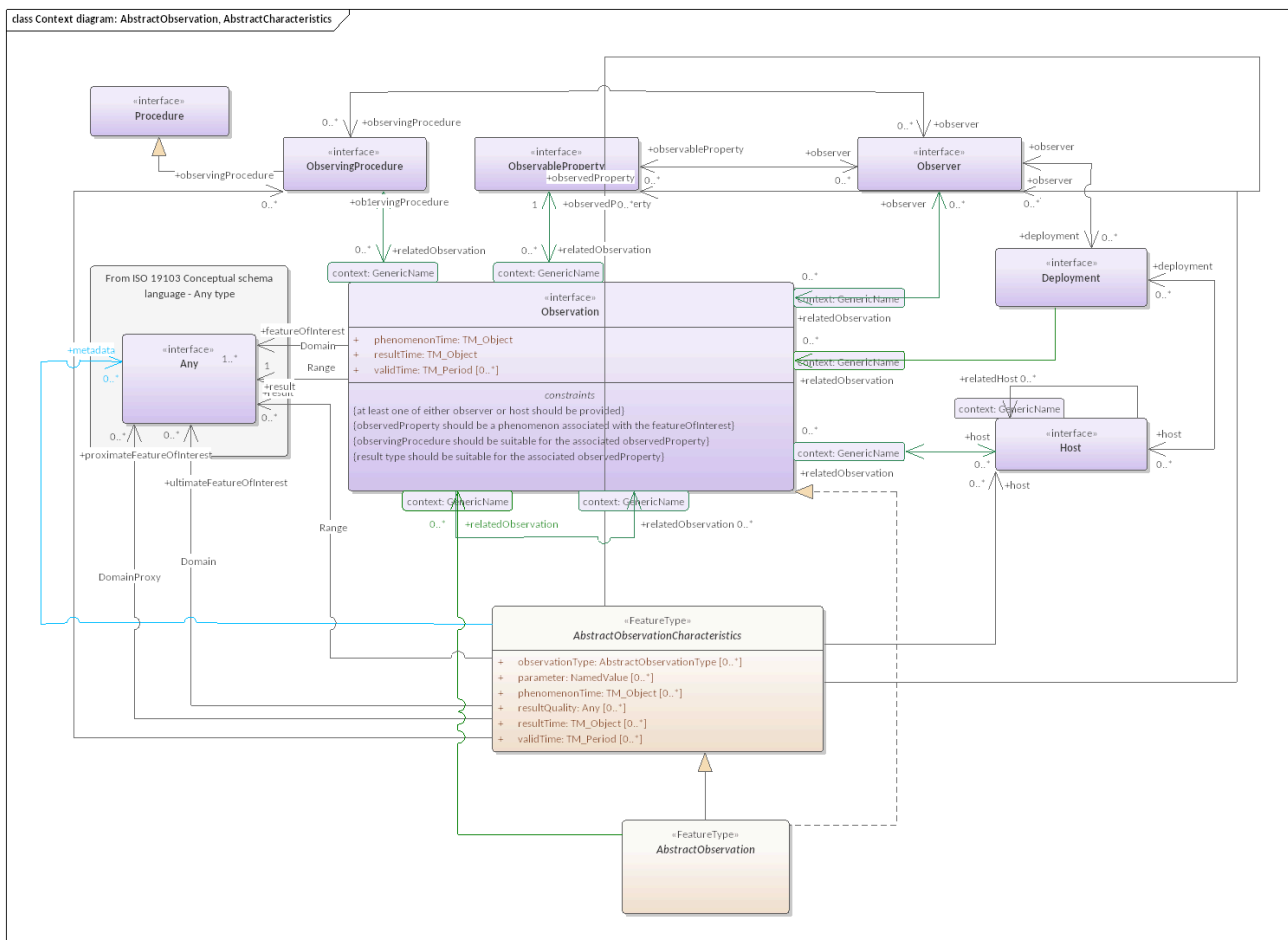


Figure 6 — Context diagram: AbstractObservation, AbstractCharacteristics

5.2 Defining tables

Table 1 — Elements of “Abstract Observation core::AbstractDeployment” (FeatureType)

Name	AbstractDeployment		
Definition	A FeatureType realization of Conceptual Observation Schema:Deployment intended as a parent class for all Deployment realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	Deployment		
Abstract	True		
Associations	Name	Type	Definition
	Host	Host	(none)
	source:(self) [none]	AbstractDeployment	(none)
	target:host [0..*]	Host	Host to which this Deployment pertains
	Observation	Observation	(none)
	source:(self) [none]	AbstractDeployment	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the deployment
	Observer	Observer	(none)
	source:(self) [none]	AbstractDeployment	(none)
	target:observer [0..*]	Observer	Observer associated with this Deployment
	Any	Any	(none)
	source:(self) [none]	AbstractDeployment	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
Attributes	Name	Type	Definition
	deploymentReason [*]	CharacterString	a human readable description of the reason for the Deployment. EXAMPLES — A researcher involved in a biodiversity survey campaign assessing the distribution of selected alien species. The deploymentReason describes the fact that this individual was involved in this campaign for the reason of identifying alien species. — A sensor is mounted on a building to monitor seismic activities A new sensor type is rolled out within a regional or thematic network due to new legal reporting requirements.
	deploymentTime [*]	TM_Object	time that the Deployment pertains to. EXAMPLES — A researcher involved in a biodiversity survey campaign assessing the distribution of selected alien species. The deploymentTime provides the time period(s) during which this person carried

Table 1 (continued)

			out this activity in the framework of the campaign. A sensor is mounted on a building to monitor seismic activities. The deploymentTime provides the time period(s) during which this sensor is mounted or active.
Constraints	(none)		

Table 2 — Elements of “Abstract Observation core::AbstractHost” (FeatureType)

Name	AbstractHost		
Definition	A FeatureType realization of Conceptual Observation Schema:Host		
Stereotype	FeatureType		
Generalization of	Host		
Abstract	True		
Associations	Name	Type	Definition
	Any	Any	(none)
	source:(self) [none]	AbstractHost	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Deployment	Deployment	(none)
	source:(self) [none]	AbstractHost	(none)
	target:deployment [0..*]	Deployment	Deployment at this Host
	Host	Host	(none)
	source:(self) [none]	AbstractHost	(none)
	target:relatedHost [0..*]	Host	Another Host the context Host is related to
	Observation	Observation	(none)
	source:(self) [none]	AbstractHost	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the host
Attributes	(none)		
Constraints	(none)		

Table 3 — Elements of “Abstract Observation core::AbstractObservableProperty” (FeatureType)

Name	AbstractObservableProperty
Definition	A FeatureType realization of Conceptual Observation Schema:ObservableProperty intended as a parent class for all ObservableProperty realizations based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Generalization of	ObservableProperty
Abstract	True

Table 3 (continued)

Associations	Name	Type	Definition
	Observer	Observer	(none)
	source:(self) [none]	AbstractObservableProperty	(none)
	target:observer [0..*]	Observer	Observer capable of observing this ObservableProperty
	Any	Any	(none)
	source:(self) [none]	AbstractObservableProperty	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Observation	Observation	(none)
	source:(self) [none]	AbstractObservableProperty	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the observable property
Attributes	(none)		
Constraints	(none)		

Table 4 — Elements of “Abstract Observation core::AbstractObservation” (FeatureType)

Name	AbstractObservation		
Definition	A FeatureType realization of Conceptual Observation Schema:Observation intended as a parent class for all Observation realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractObservationCharacteristics		
Generalization of	Observation		
Abstract	True		
Associations	Name	Type	Definition
	Observation	Observation	(none)
	source:(self) [none]	AbstractObservation	(none)
	target:relatedObservation [0..*]	Observation	Another Observation related to the context Observation
Attributes	(none)		
Constraints	(none)		

Table 5 — Elements of “Abstract Observation core::AbstractObservationCharacteristics” (FeatureType)

Name	AbstractObservationCharacteristics		
Definition	set of common characteristics used for describing an Observation or a collection of Observations.		
Stereotype	FeatureType		
Generalization of	AbstractObservation, ObservationCharacteristics		
Abstract	True		

Table 5 (continued)

Associations	Name	Type	Definition
	DomainProxy	Any	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target: proximateFeatureOfInterest [0..*]	Any	entity that is directly of interest in the act of observing captured by the Observation or a collection of Observations described.
	Host	Host	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:host [0..*]	Host	Host that is involved in the creation of the Observation or a collection of Observations described
	Any	Any	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Domain	Any	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target: ultimateFeatureOfInterest [0..*]	Any	entity that is ultimately of interest in the act of observing captured by the Observation or a collection of Observations described.
	ObservingProcedure	ObservingProcedure	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:observingProcedure [0..*]	ObservingProcedure	ObservingProcedure used by the Observation or a collection of Observations described to determine the value of the ObservableProperty provided by the result.
	ObservableProperty	ObservableProperty	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:observedProperty [0..*]	ObservableProperty	ObservableProperty that is the subject of the Observation or a collection of Observations described
	Observer	Observer	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:observer [0..*]	Observer	Observer that is involved in the creation of Observation or a collection of Observations described

Table 5 (continued)

	Range	Any	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:result [0..*]	Any	result of the Observation or a collection of Observations described
Attributes	Name	Type	Definition
	observationType [*]	AbstractObservationType	Information providing further detail on the type of Observations being described.
	parameter [*]	NamedValue	arbitrary event-specific parameter relevant to one or more Observations. NOTE This might be an environmental parameter, an instrument setting or input, or an event-specific sampling parameter that is not tightly bound to either the feature-of-interest or to the observation procedure. NOTE Parameters that are tightly bound to the procedure can be recorded as part of the procedure description. The AbstractObservingProcedure is a generic or standard procedure, rather than an event-specific process. In this context, parameters bound to the observation act, such as instrument settings, calibrations or inputs, local position, detection limits, asset identifier, operator, may augment the description of a standard procedure. EXAMPLE A time sequence of observations of water quality in a well might be made at variable depths within the well. While these can be associated with specimens taken from the well at this depth as the features-of-interest, a more common approach is to identify the well itself as the feature-of-interest, and add a ?samplingDepth? parameter to the observation. The sampling depth is of secondary interest compared to the temporal variation of water quality at the site.

Table 5 (continued)

phenomenonTime [*]	TM_Object	time that the Result applies to the characteristic of the FeatureOfInterest being observed. NOTE 1 The phenomenonTime is often the time the Sample has been taken. NOTE 2 This is often the time of interaction by a sampling Procedure or observation Procedure with a real-world feature. NOTE 3 If the result is the average of multiple samples taken at different times, then the phenomenonTime is the time interval over which these measurements were taken.
resultQuality [*]	Any	Information pertaining to the data quality of the result
resultTime [*]	TM_Object	instant of time when the result of the Observation became available. EXAMPLE 1 The resultTime typically corresponds to when the Procedure associated with the Observation was completed. For some observations this is identical to the phenomenonTime. However, there are important cases where they differ. EXAMPLE 2 Where a measurement is made on a specimen in a laboratory, the phenomenonTime is the time the specimen was retrieved from its host, while the resultTime is the time the laboratory procedure was applied. EXAMPLE 3 The resultTime also supports disambiguation of repeat measurements made of the same property of a feature using the same procedure. EXAMPLE 4 Where sensor observation results are post-processed, the resultTime is the post-processing time, while the phenomenonTime is the time of initial interaction with the world. EXAMPLE 5 Simulations may be used to estimate the values for phenomena in the future or past. The phenomenonTime is the time that the result

Table 5 (continued)

			applies to, while the resultTime is the time that the simulation was executed.
	validTime [*]	TM_Period	The time interval during which the result is assumed to be applicable for use. NOTE This attribute is commonly required in forecasting applications.
Constraints	(none)		

Table 6 — Elements of “Abstract Observation core::AbstractObservationType” (CodeList)

Name	AbstractObservationType
Definition	an empty codelist parent class for Observation classification schemes. All codelists used for classifying Observations SHALL be specialized from this class.
Stereotype	CodeList
Generalization of	ObservationTypeByResultType
Abstract	True
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 7 — Elements of “Abstract Observation core::AbstractObserver” (FeatureType)

Name	AbstractObserver		
Definition	A FeatureType realization of Conceptual Observation Schema:Observer intended as a parent class for all Observer realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	Observer		
Abstract	True		
Associations	Name	Type	Definition
	Any	Any	(none)
	source:(self) [none]	AbstractObserver	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	ObservingProcedure	ObservingProcedure	(none)
	source:(self) [none]	AbstractObserver	(none)
	target:observingProcedure [0..*]	ObservingProcedure	An ObservingProcedure that this Observer can perform
	ObservableProperty	ObservableProperty	(none)
	source:(self) [none]	AbstractObserver	(none)
	target:observableProperty [0..*]	ObservableProperty	ObservableProperty that this Observer can observe

Table 7 (continued)

	Deployment	Deployment	(none)
	source:(self) [none]	AbstractObserver	(none)
	target:deployment [0..*]	Deployment	Deployment to which this Observer is either physically or organizationally attached.
	Observation	Observation	(none)
	source:(self) [none]	AbstractObserver	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the observer
Attributes	(none)		
Constraints	(none)		

Table 8 — Elements of “Abstract Observation core::AbstractObservingProcedure” (FeatureType)

Name	AbstractObservingProcedure		
Definition	A FeatureType realization of Conceptual Observation Schema:ObservingProcedure intended as a parent class for all :ObservingProcedure realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	ObservingProcedure		
Abstract	True		
Associations	Name	Type	Definition
	Observation	Observation	(none)
	source:(self) [none]	AbstractObservingProcedure	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the observing procedure
	Observer	Observer	(none)
	source:(self) [none]	AbstractObservingProcedure	(none)
	target:observer [0..*]	Observer	Observer capable of performing this ObservingProcedure.
	Any	Any	(none)
	source:(self) [none]	AbstractObservingProcedure	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
Attributes	(none)		
Constraints	(none)		

Table 9 — Elements of “Abstract Observation core::NamedValue” (DataType)

Name	NamedValue
Definition	The class <i>NamedValue</i> provides for a generic soft-typed parameter value. NamedValue shall support two attributes.
Stereotype	DataType

Table 9 (continued)

Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	name [1]	GenericName	the meaning of the named value. Value should be taken from a well-governed source if possible. EXAMPLES When used as the value of an Observation::parameter, the name might take values like ?procedureOperator?, ?detectionLimit?, ?amplifierGain?, ?samplingDepth?.
	value [1]	Any	the value of the parameter. The type Any should be substituted by a suitable concrete type, such as CI_ResponsibleParty or Measure.
Constraints	(none)		

Table 10 — Elements of “Abstract Observation core::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 11 — Elements of “Abstract Observation core::From ISO 19156 Observations and measurements — Conceptual observation schema” ()

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 12 — Elements of “Abstract Observation core::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 13 — Elements of “Abstract Observation core::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	

Table 13 (*continued*)

Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 14 — Elements of “Abstract Observation core::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 15 — Elements of “Abstract Observation core::From ISO 19156 Observations and measurements — Conceptual observation schema” ()

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 16 — Elements of “Abstract Observation core::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

6 Abstract Sample core package

6.1 Abstract Sample core overview

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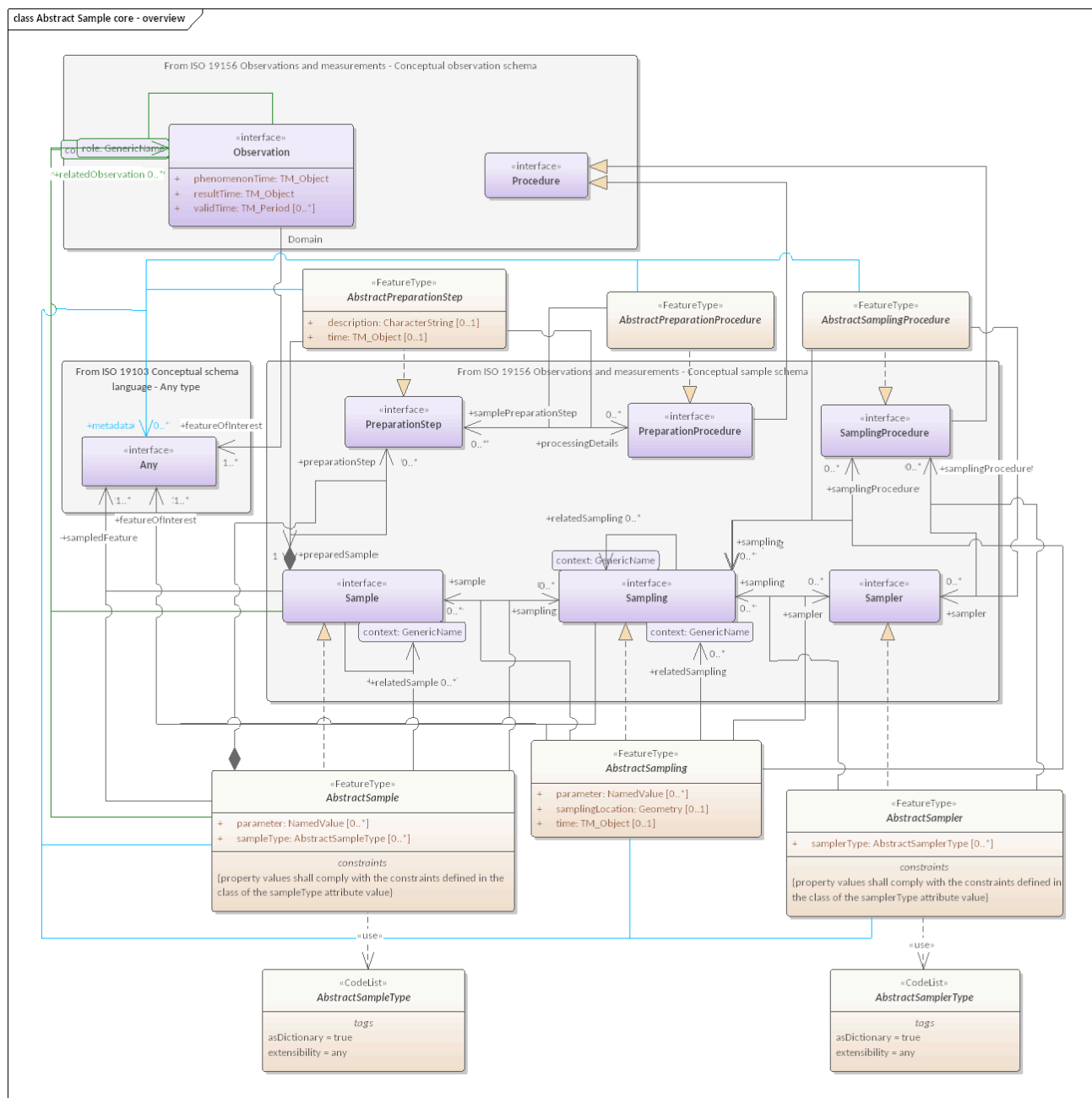


Figure 7 — Abstract Sample core - overview

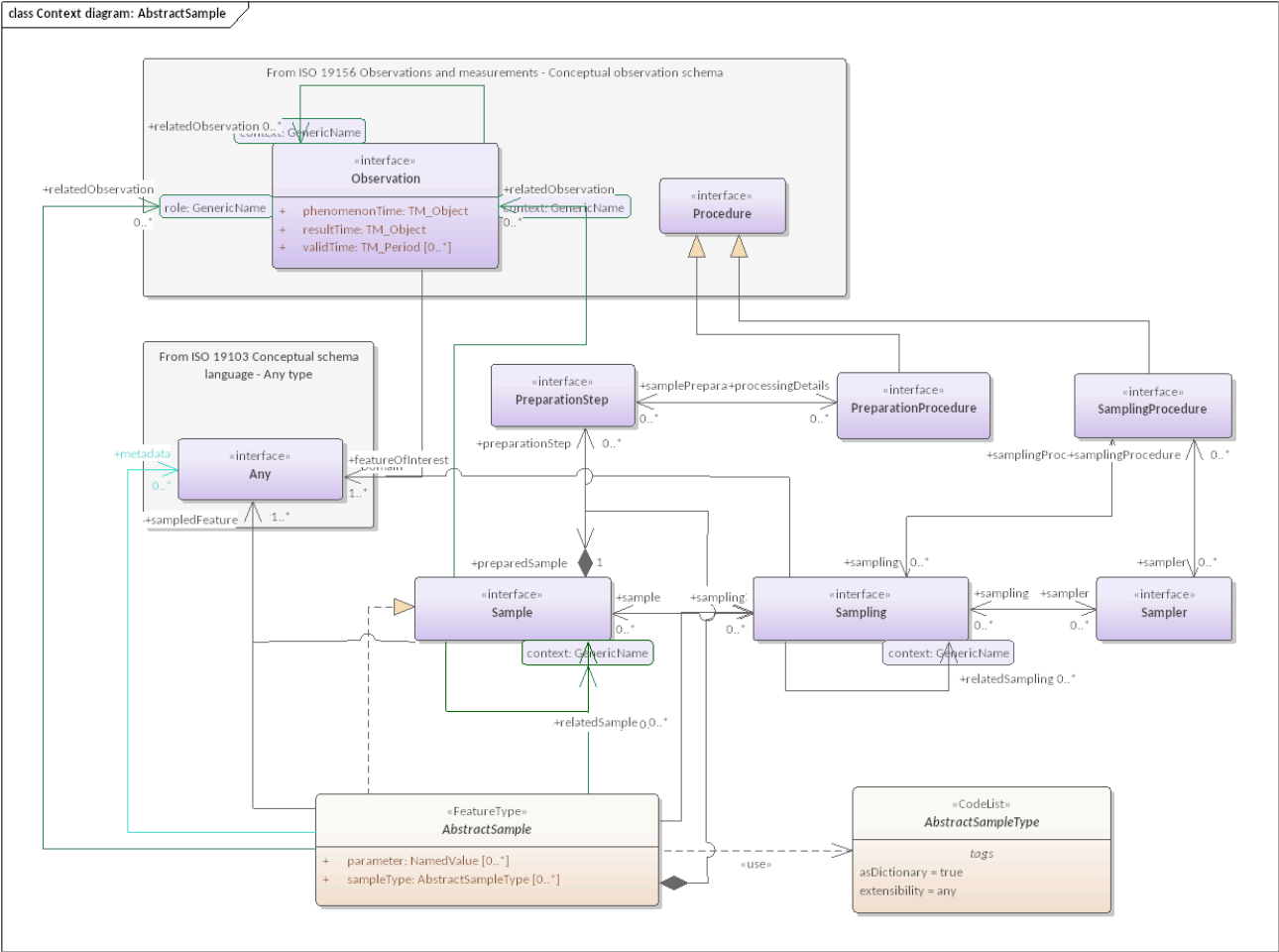


Figure 8 — Context diagram: AbstractSample

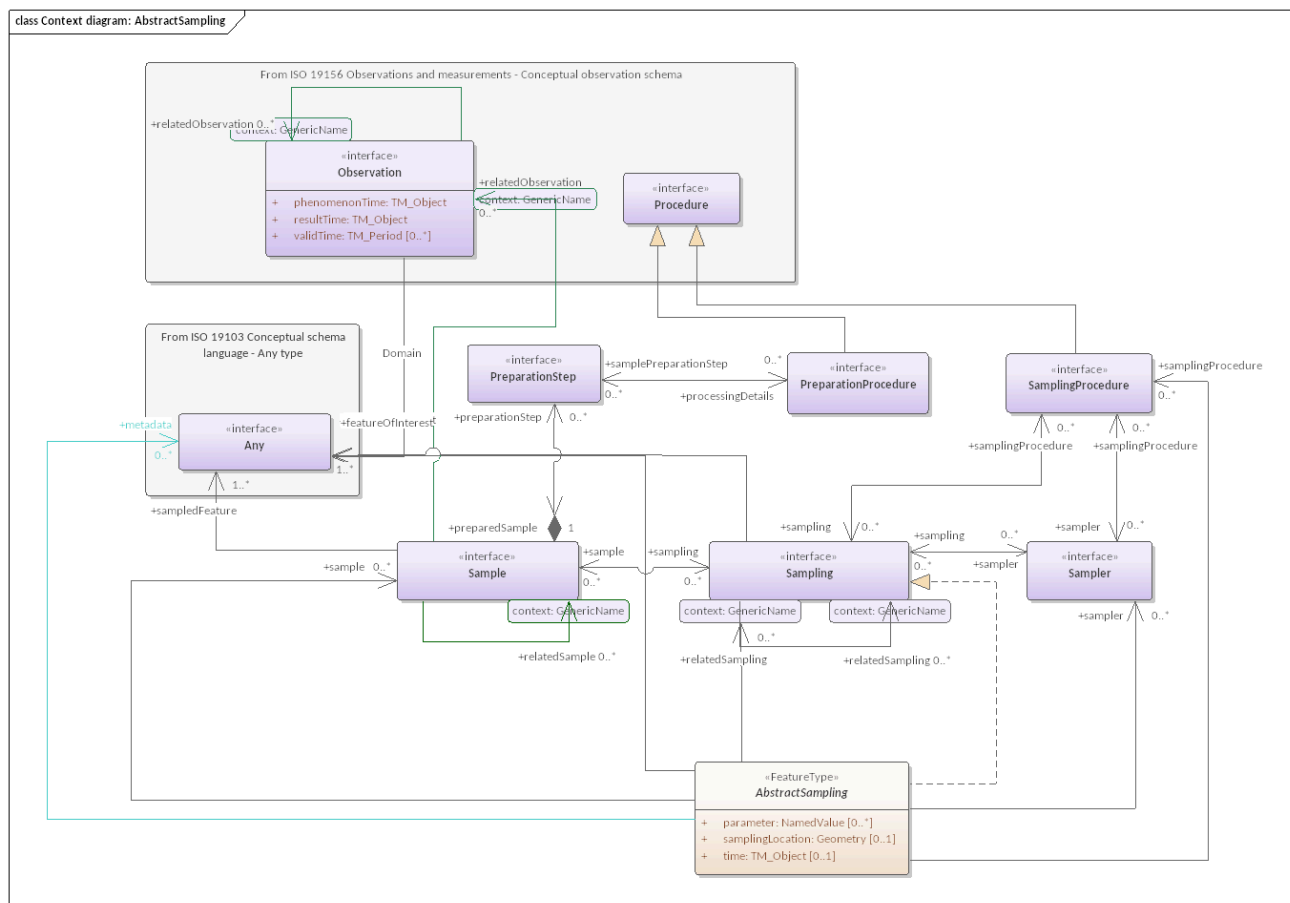


Figure 9 — Context diagram: AbstractSampling

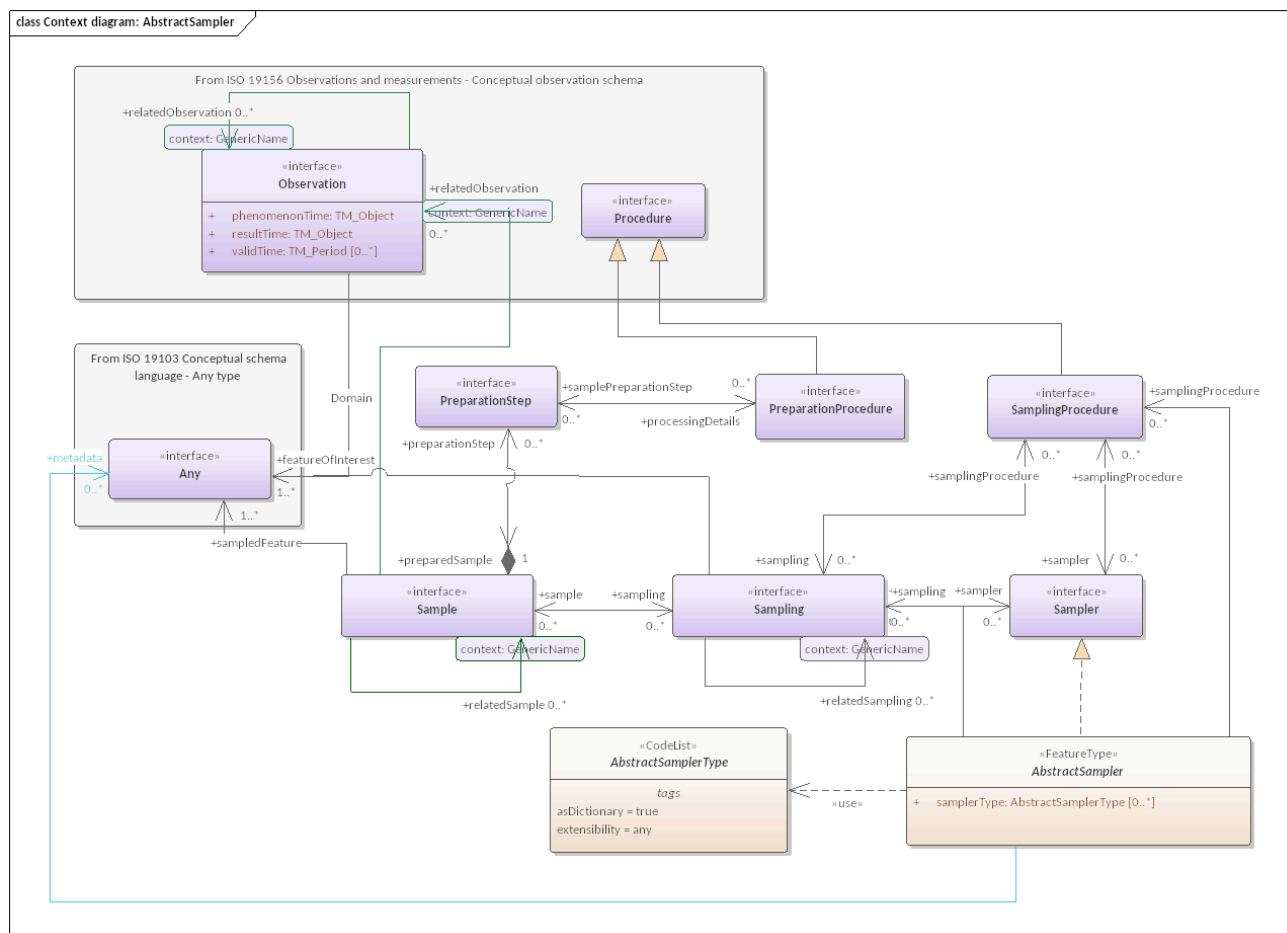


Figure 10 — Context diagram: AbstractSampler

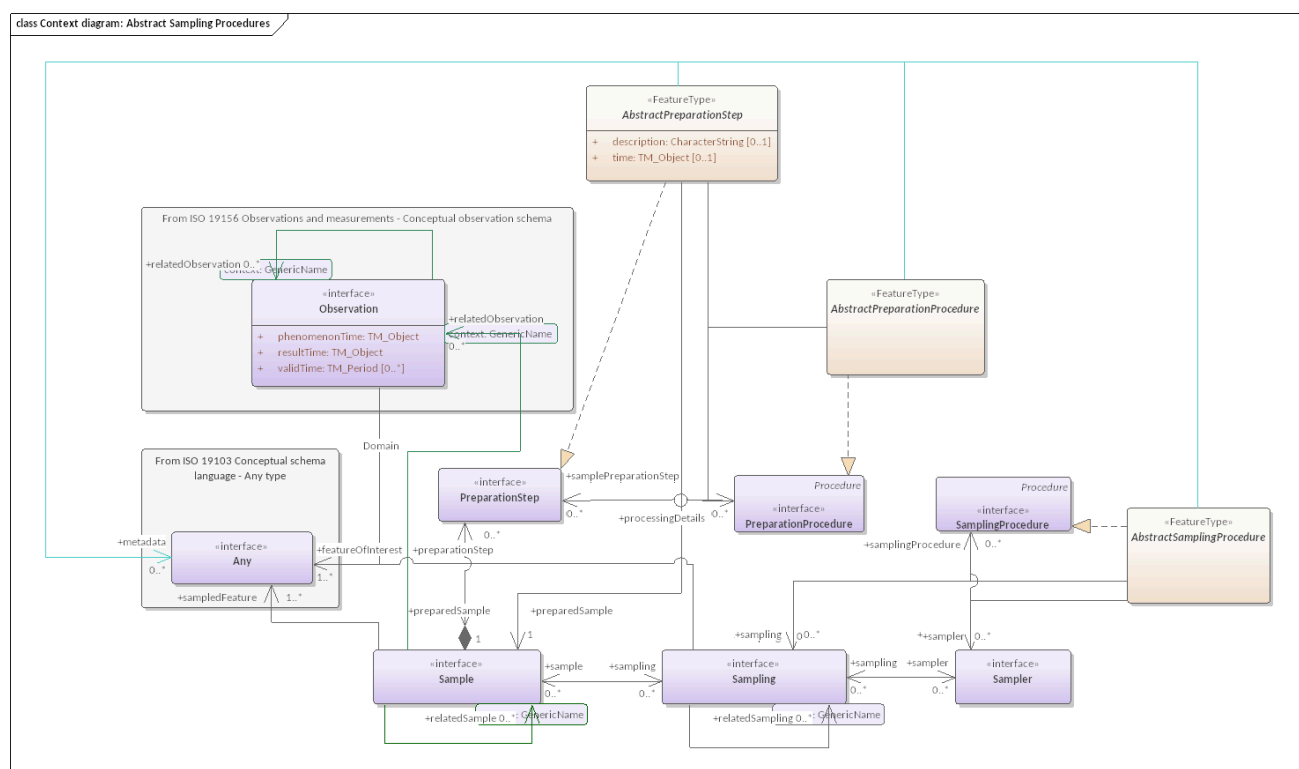


Figure 11 — Context diagram: Abstract Sampling Procedures

6.2 Defining tables

Table 17 — Elements of “Abstract Sample core::AbstractPreparationProcedure” (FeatureType)

Name	AbstractPreparationProcedure		
Definition	A FeatureType realization of Conceptual Sample Schema:PreparationProcedure intended as a parent class for all PreparationProcedure realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	PreparationProcedure		
Abstract	True		
Associations	Name	Type	Definition
	Any	Any	(none)
	source:(self) [none]	AbstractPreparationProcedure	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	PreparationStep	PreparationStep	(none)
	source:(self) [none]	AbstractPreparationProcedure	(none)
	target:samplePreparationStep [0..*]	PreparationStep	provides information on the PreparationSteps where this procedure has been used
Attributes	(none)		
Constraints	(none)		

Table 18 — Elements of “Abstract Sample core::AbstractPreparationStep” (FeatureType)

Name	AbstractPreparationStep		
Definition	A FeatureType realization of Conceptual Sample Schema:PreparationStep intended as a parent class for all PreparationStep realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	PreparationStep		
Abstract	True		
Associations	Name	Type	Definition
	Sample	Sample	(none)
	source:(self) [none]	AbstractPreparationStep	(none)
	target:preparedSample [1]	Sample	Sample on which the PreparationProcedure is performed
	PreparationProcedure	PreparationProcedure	(none)
	source:(self) [none]	AbstractPreparationStep	(none)
	target:processingDetails [0..*]	PreparationProcedure	PreparationProcedure step performed on the Sample the PreparationStep pertains to
	Any	Any	(none)
	source:(self) [none]	AbstractPreparationStep	(none)

Table 18 (continued)

	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
Attributes	Name	Type	Definition
	description [0..1]	CharacterString	description of the preparationStep
	time [0..1]	TM_Object	time of the preparationStep
Constraints	(none)		

Table 19 — Elements of “Abstract Sample core::AbstractSample” (FeatureType)

Name	AbstractSample		
Definition	A FeatureType realization of Conceptual Sample Schema:Sample intended as a parent class for all Sample realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	Sample		
Abstract	True		
Associations	Name	Type	Definition
	PreparationStep	PreparationStep	(none)
	source:(self) [none]	AbstractSample	(none)
	target:preparationStep [0..*]	PreparationStep	PreparationStep(s) applied to prepare the Sample
	Observation	Observation	(none)
	source:(self) [none]	AbstractSample	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the sample
	Sampling	Sampling	(none)
	source:(self) [none]	AbstractSample	(none)
	target:sampling [0..*]	Sampling	Sampling the Sample is the result of
	Any	Any	(none)
	source:(self) [none]	AbstractSample	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Any	Any	(none)
	source:(self) [none]	AbstractSample	(none)
	target:sampledFeature [1..*]	Any	the feature the Sample is intended to be representative of. NOTE The sampled feature is usually a real-world feature from an application domain.
	Sample	Sample	(none)
	source:(self) [none]	AbstractSample	(none)

Table 19 (continued)

	target:relatedSample [0..*]	Sample	Another Sample the context Sample is related to
Attributes	Name	Type	Definition
	parameter [*]	NamedValue	arbitrary event-specific parameter relevant to the Sample
	sampleType [*]	AbstractSampleType	type of Sample according to a community agreed typology.
Constraints	(none)		

Table 20 — Elements of “Abstract Sample core::AbstractSampler” (FeatureType)

Name	AbstractSampler		
Definition	A FeatureType realization of Conceptual Sample Schema:Sampler intended as a parent class for all Sampler realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	Sampler		
Abstract	True		
Associations	Name	Type	Definition
	SamplingProcedure	SamplingProcedure	(none)
	source:(self) [none]	AbstractSampler	(none)
	target: samplingProcedure [0..*]	SamplingProcedure	Procedure implemented by the Sampler
	Any	Any	(none)
	source:(self) [none]	AbstractSampler	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Sampling	Sampling	(none)
	source:(self) [none]	AbstractSampler	(none)
	target:sampling [0..*]	Sampling	Sampling act performed by the Sampler
Attributes	Name	Type	Definition
	samplerType [*]	AbstractSamplerType	type of Sampler according to a community agreed typology
Constraints	(none)		

Table 21 — Elements of “Abstract Sample core::AbstractSamplerType” (CodeList)

Name	AbstractSamplerType
Definition	an empty codelist parent class for Sampler classification schemes. All codelists used for classifying Samplers SHALL be specialized from this class.
Stereotype	CodeList
Generalization of	SamplerClassification
Abstract	True
Associations	(none)

Table 21 (continued)

Attributes	(none)
Constraints	(none)

Table 22 — Elements of “Abstract Sample core::AbstractSampleType” (CodeList)

Name	AbstractSampleType
Definition	an empty codelist parent class for Sample classification schemes. All codelists used for classifying Samples SHALL be specialized from this class.
Stereotype	CodeList
Generalization of	SampleTypeByGeometryType, SampleTypeByMaterialClass
Abstract	True
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 23 — Elements of “Abstract Sample core::AbstractSampling” (FeatureType)

Name	AbstractSampling		
Definition	A FeatureType realization of Conceptual Sample Schema:Sampling intended as a parent class for all Sampling realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	Sampling		
Abstract	True		
Associations	Name	Type	Definition
	SamplingProcedure	SamplingProcedure	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:samplingProcedure [0..*]	SamplingProcedure	SamplingProcedure used by the Sampling
	Any	Any	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Sampler	Sampler	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:sampler [0..*]	Sampler	Sampler that performed the Sampling
	Sampling	Sampling	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:relatedSampling [0..*]	Sampling	Another Sampling the context Sampling is related to
	Any	Any	(none)
	source:(self) [none]	AbstractSampling	(none)

Table 23 (continued)

	target:featureOfInterest [1..*]	Any	The concept, real-world object or phenomenon (feature-of-interest) the Sample(s) of the Sampling represent
	Sample	Sample	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:sample [0..*]	Sample	Sample generated by the Sampling
Attributes	Name	Type	Definition
	parameter [*]	NamedValue	arbitrary event-specific parameter relevant to the Sampling
	samplingLocation [0..1]	Geometry	location information pertaining to the Sampling
	time [0..1]	TM_Object	time of the Sampling
Constraints	(none)		

Table 24 — Elements of “Abstract Sample core::AbstractSamplingProcedure” (FeatureType)

Name	AbstractSamplingProcedure		
Definition	A FeatureType realization of Conceptual Sample Schema:SamplingProcedure intended as a parent class for all SamplingProcedure realizations based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Generalization of	SamplingProcedure		
Abstract	True		
Associations	Name	Type	Definition
	Any	Any	(none)
	source:(self) [none]	AbstractSamplingProcedure	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Sampling	Sampling	(none)
	source:(self) [none]	AbstractSamplingProcedure	(none)
	target:sampling [0..*]	Sampling	provides information on the Sampling where this procedure has been used
	Sampler	Sampler	(none)
	source:(self) [none]	AbstractSamplingProcedure	(none)
	target:sampler [0..*]	Sampler	provides information on the Sampler that implements this procedure
Attributes	(none)		
Constraints	(none)		

Table 25 — Elements of “Abstract Sample core::From ISO 19156 Observations and measurements — Conceptual sample schema” ()

Name	From ISO 19156 Observations and measurements — Conceptual sample schema
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7 Basic Observations package

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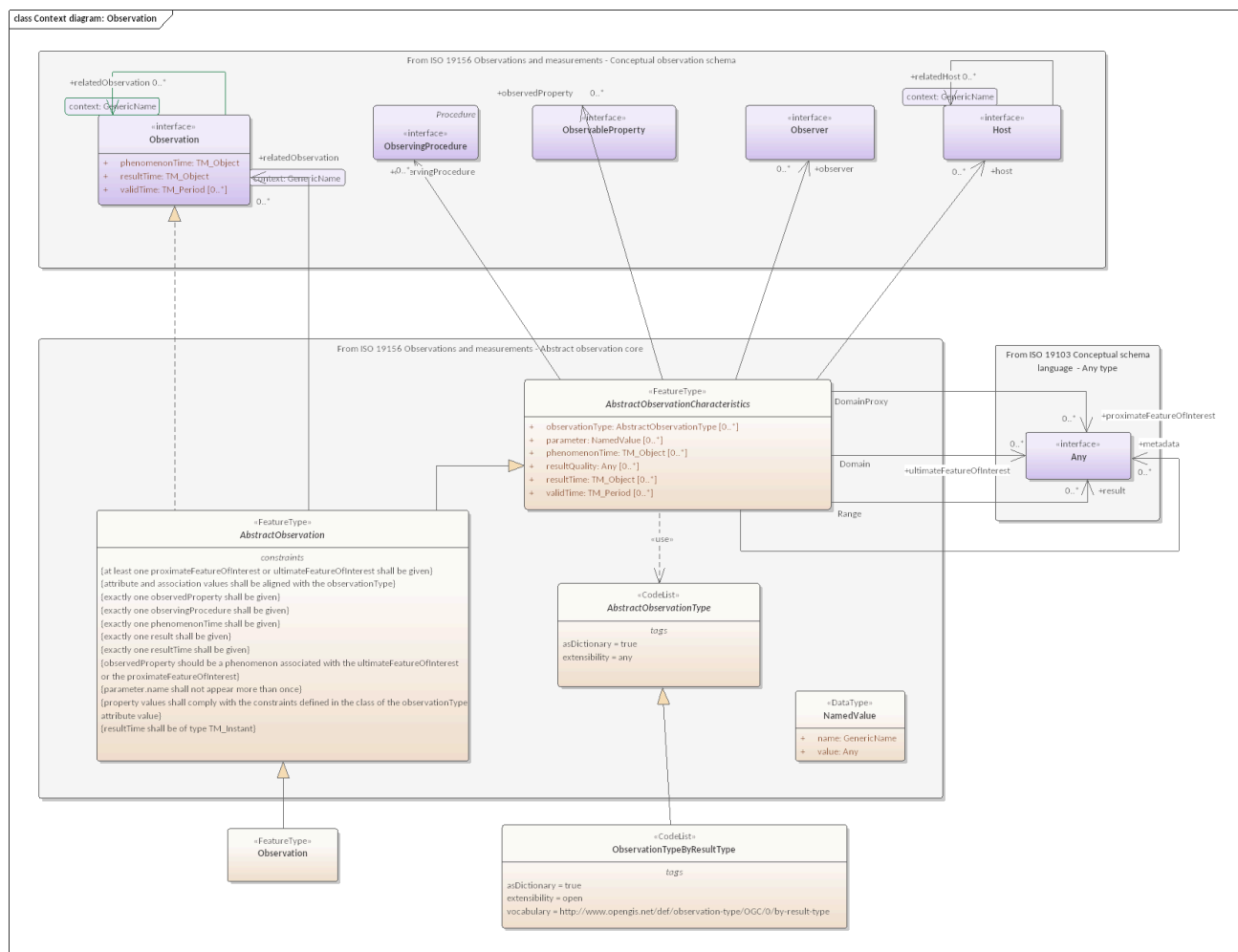


Figure 13 — Context diagram: Observation

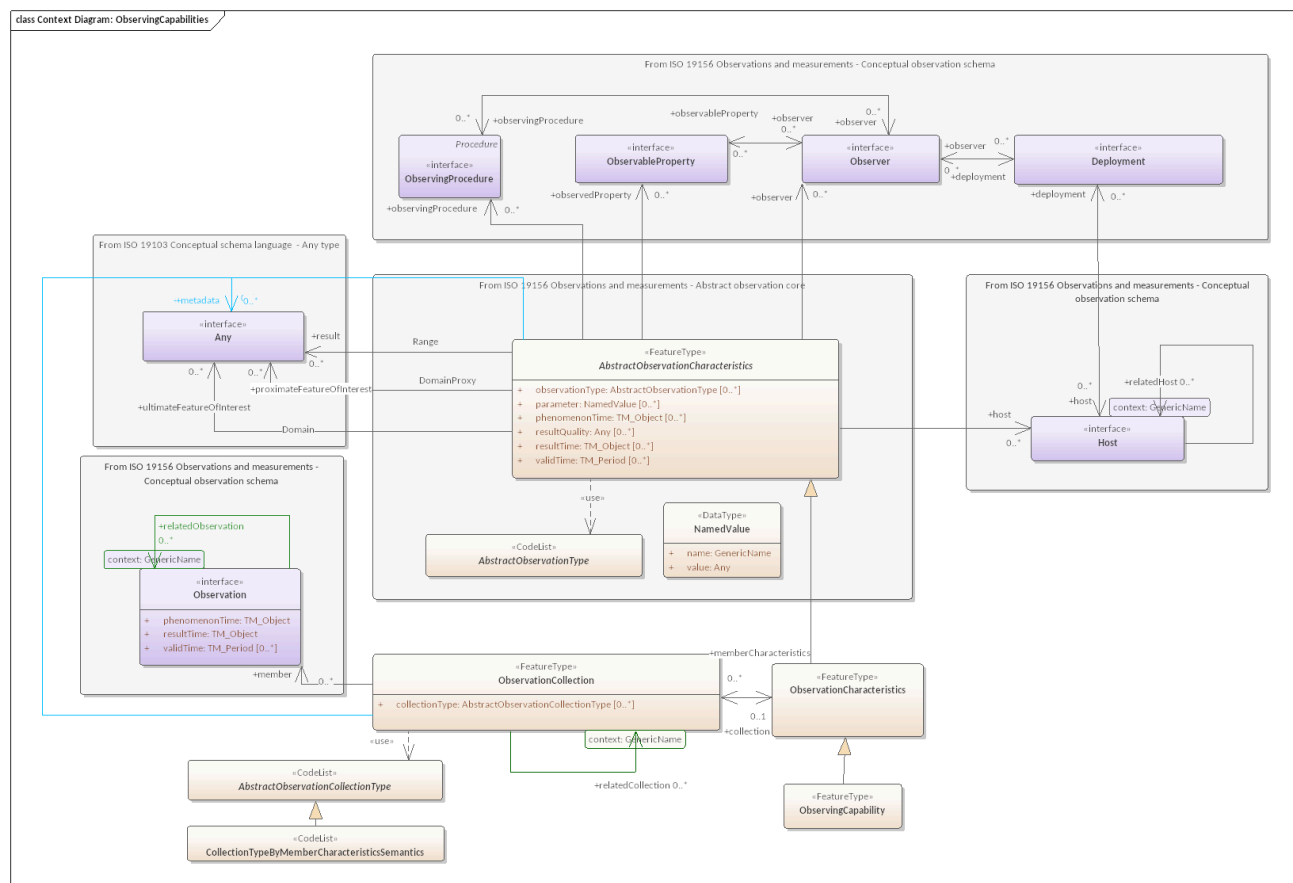


Figure 14 — Context Diagram: ObservingCapabilities

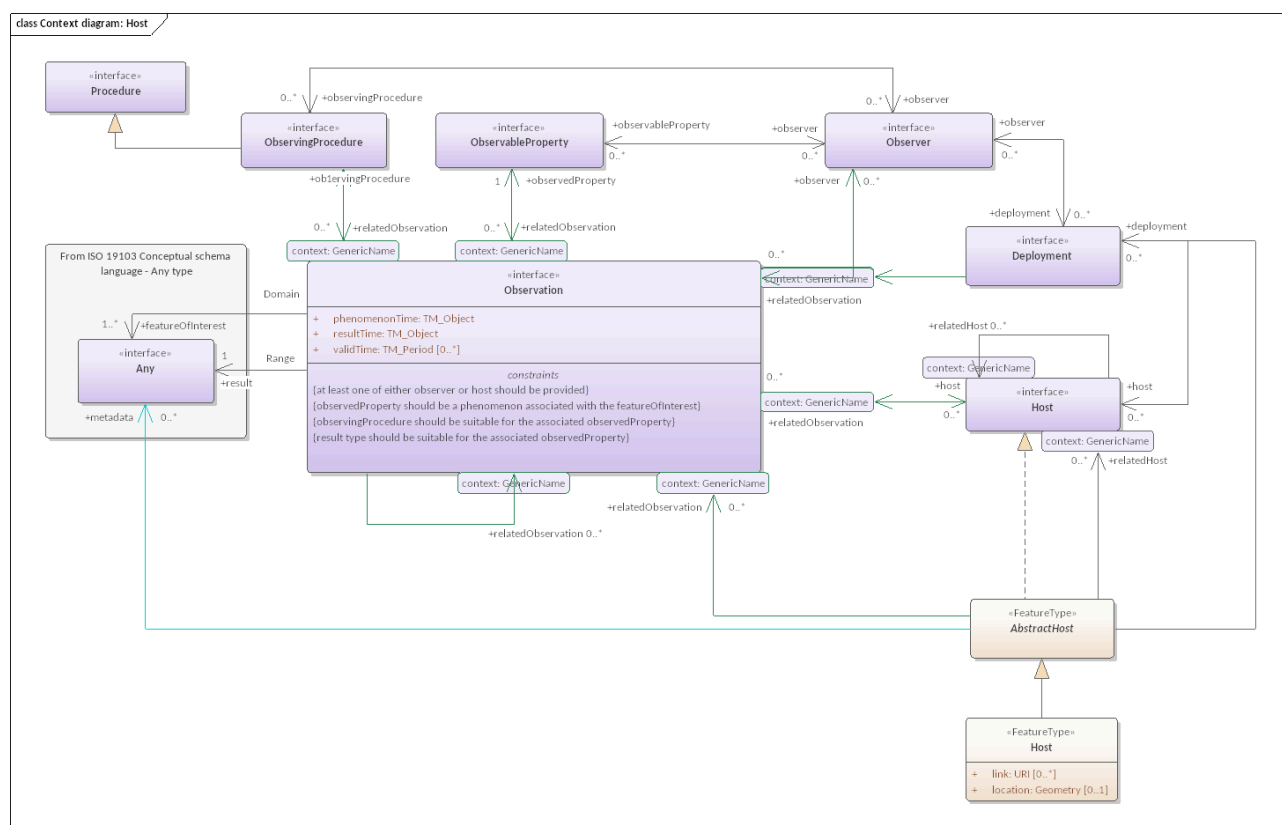


Figure 15 — Context diagram: Host

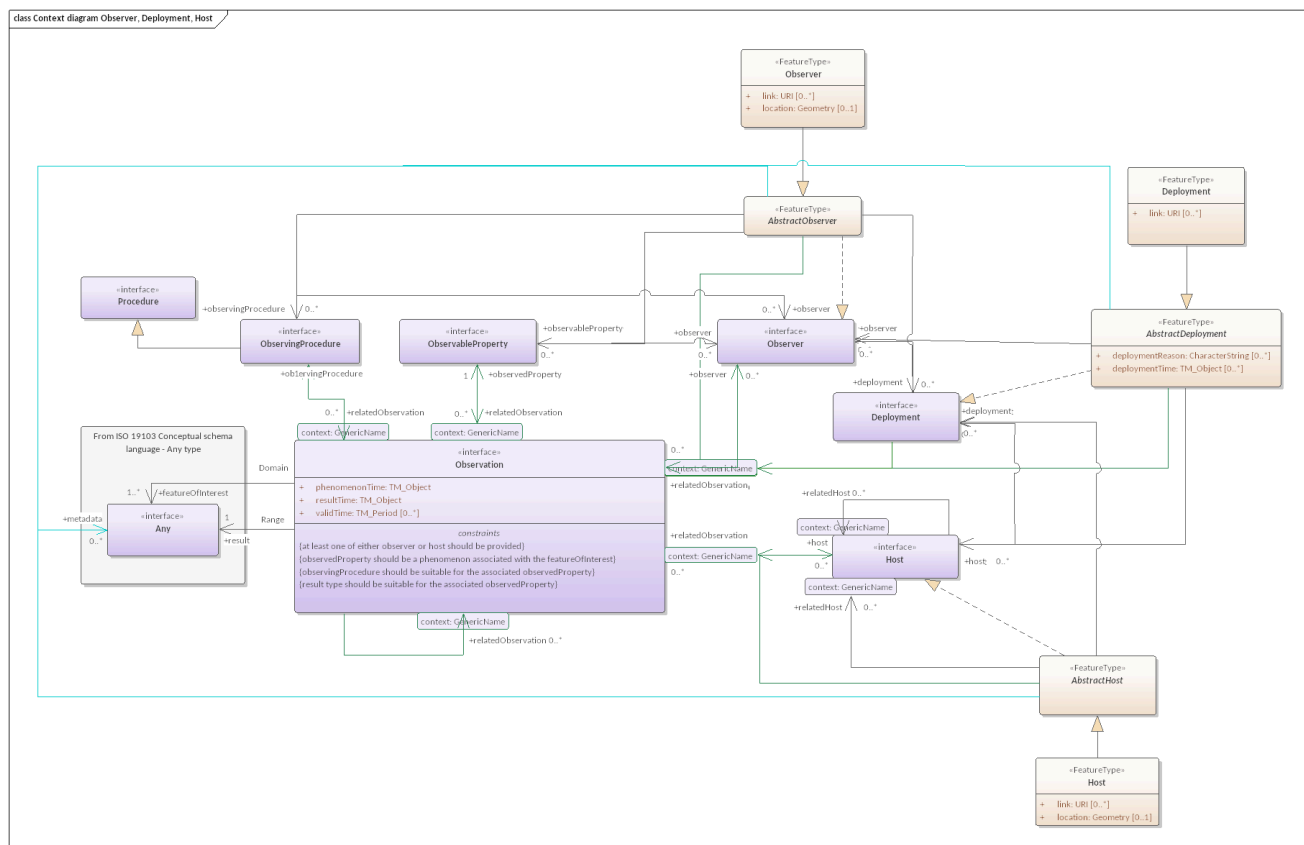


Figure 16 — Context diagram Observer, Deployment, Host

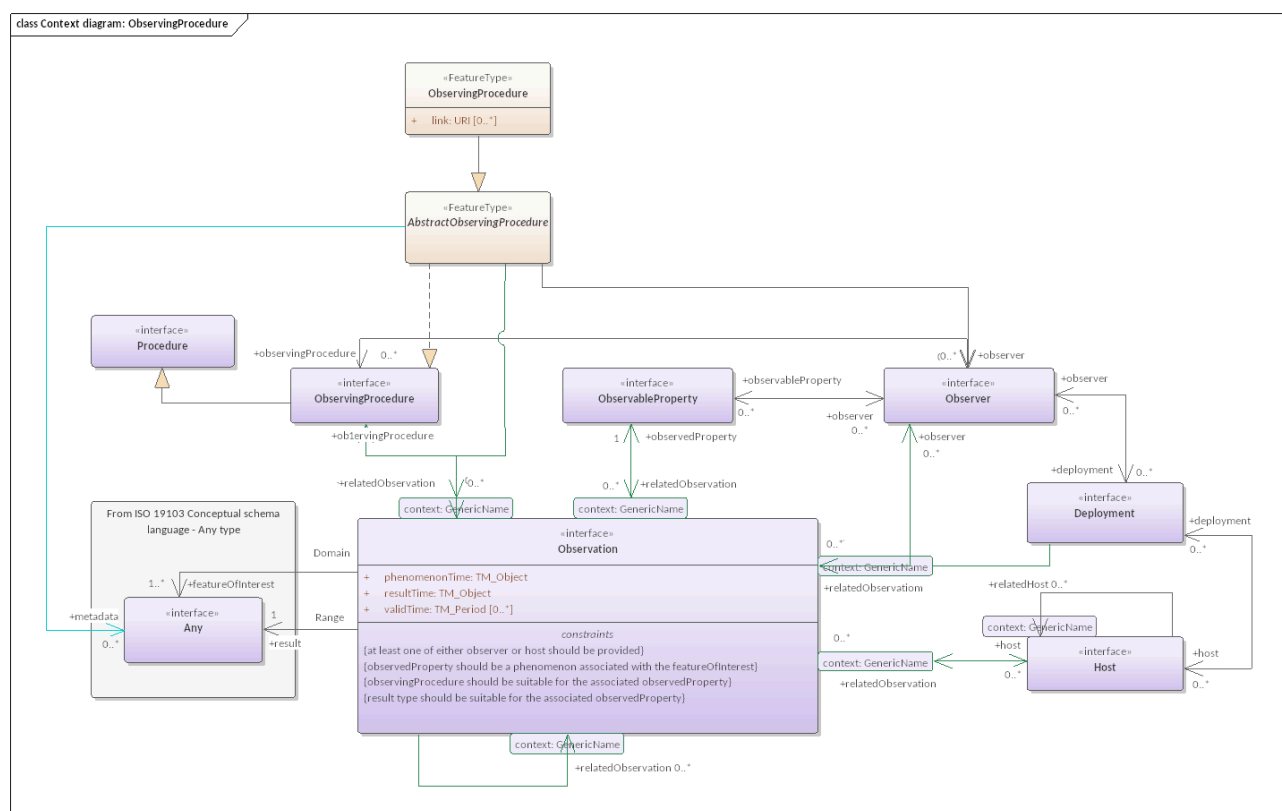


Figure 17 — Context diagram: ObservingProcedure

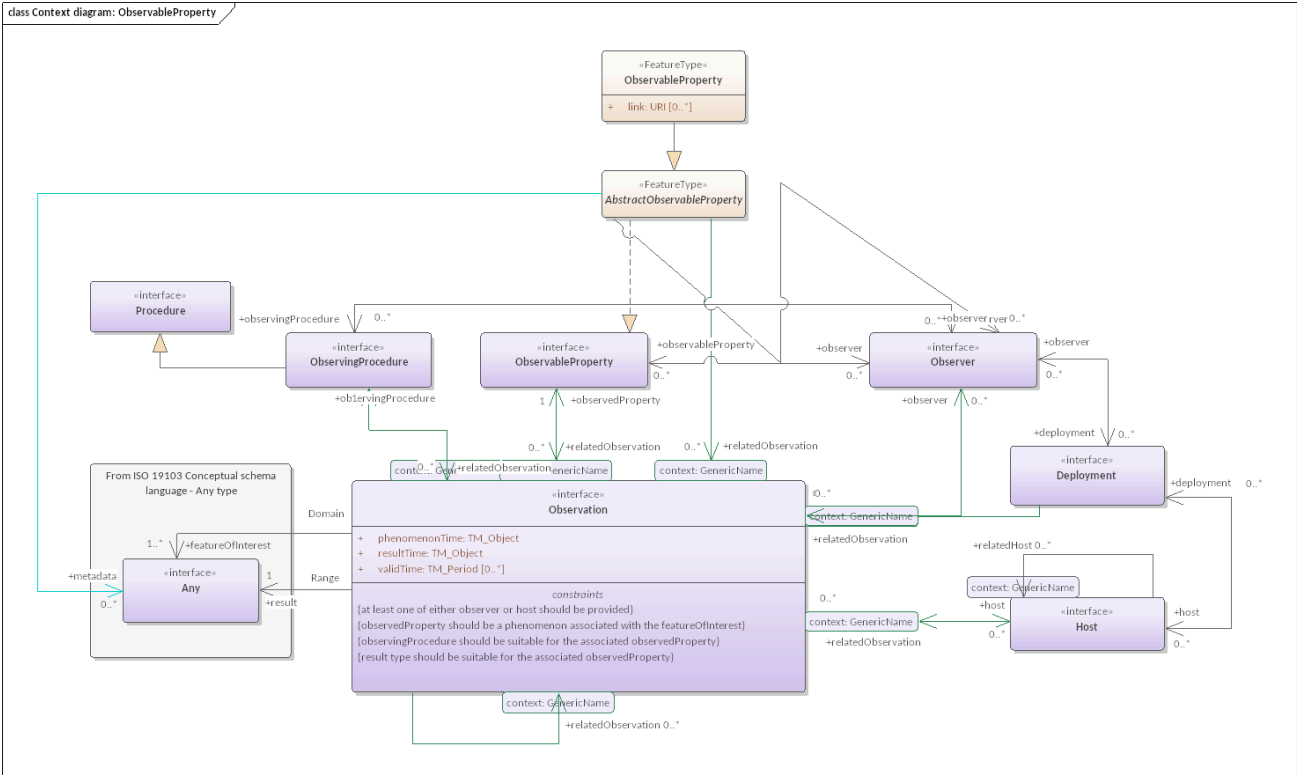


Figure 18 — Context diagram: ObservableProperty

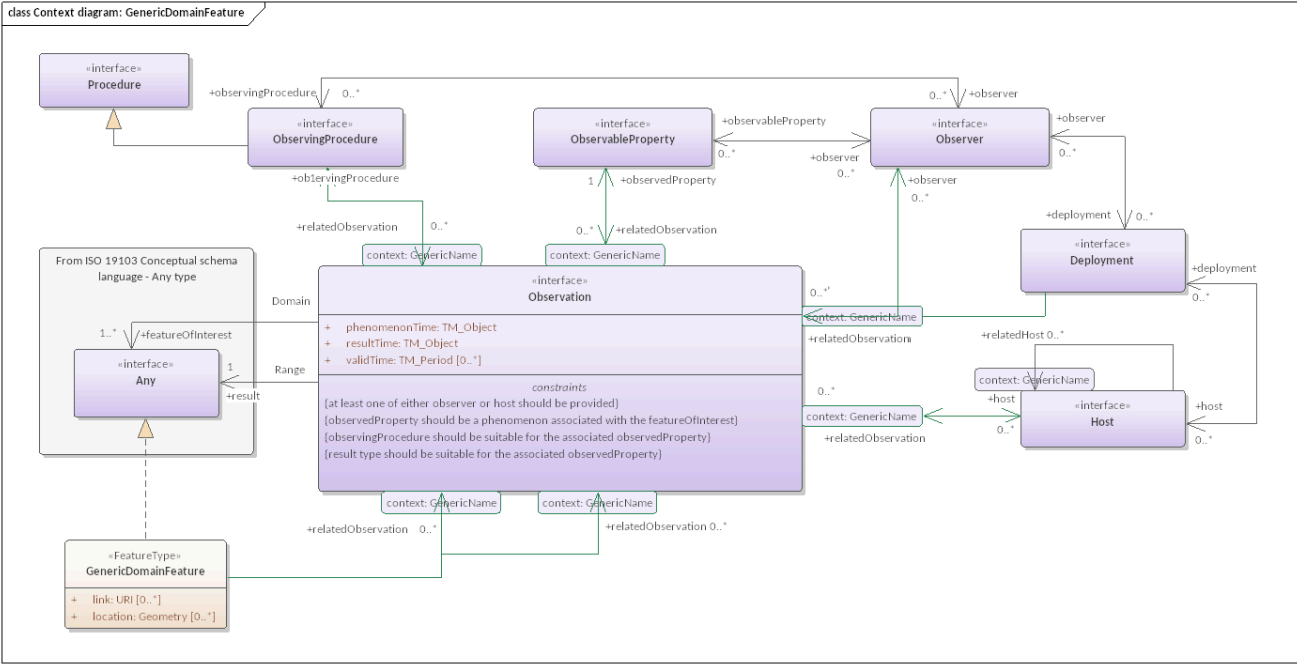


Figure 19 — Context diagram: GenericDomainFeature

7.2 Defining tables

Table 26 — Elements of “Basic Observations::AbstractObservationCollectionType” (CodeList)	
Name	AbstractObservationCollectionType
Definition	an empty codelist parent class for ObservationCollection classification schemes. All codelists used for classifying ObservationCollections SHALL be specialized from this class.

Table 26 (continued)

Stereotype	CodeList
Generalization of	CollectionTypeByMemberCharacteristicsSemantics
Abstract	True
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 27 — Elements of “Basic Observations::CollectionTypeByMemberCharacteristicsSemantics” (CodeList)

Name	CollectionTypeByMemberCharacteristicsSemantics
Definition	ObservationCollection classification scheme based on how the properties of the ObservationCharacteristics associated with role memberCharacteristics shall be related to the corresponding properties of the Observations associated with role member
Stereotype	CodeList
Inherits from	AbstractObservationCollectionType
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 28 — Elements of “Basic Observations::Deployment” (FeatureType)

Name	Deployment		
Definition	Concrete class for expressing a realization of Conceptual Observation Schema:Deployment interface based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractDeployment		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	link [*]	URI	Additional descriptive resources pertaining to a feature.
Constraints	(none)		

Table 29 — Elements of “Basic Observations::GenericDomainFeature” (FeatureType)

Name	GenericDomainFeature		
Definition	A concrete featureType to be utilized as featureOfInterest of an *Observation.*NOTE: This type is foreseen as a placeholder for specialized domain features in order to enable rapid prototyping.		
Stereotype	FeatureType		
Abstract	False		
Associations	Name	Type	Definition
	Observation	Observation	(none)
	source:(self) [none]	GenericDomainFeature	(none)

Table 29 (continued)

	target:relatedObservation [0..*]	Observation	An Observation related to the domain feature
Attributes	Name	Type	Definition
	link [*]	URI	Additional descriptive resources pertaining to a feature.
	location [*]	Geometry	Location information pertaining to a feature.
Constraints	(none)		

Table 30 — Elements of “Basic Observations::Host” (FeatureType)

Name	Host		
Definition	Concrete class for expressing a realization of Conceptual Observation Schema:Host interface based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractHost		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	link [*]	URI	Additional descriptive resources pertaining to a feature.
	location [0..1]	Geometry	Location information pertaining to a feature.
Constraints	(none)		

Table 31 — Elements of “Basic Observations::ObservableProperty” (FeatureType)

Name	ObservableProperty		
Definition	Concrete class for expressing a realization of Conceptual Observation Schema:ObservableProperty interface based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractObservableProperty		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	link [*]	URI	Additional descriptive resources pertaining to a feature.
Constraints	(none)		

Table 32 — Elements of “Basic Observations::Observation” (FeatureType)

Name	Observation		
Definition	Concrete class for expressing a realization of Conceptual Observation Schema:Observation interface based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractObservation		
Abstract	False		

Table 32 (continued)

Associations	(none)
Attributes	(none)
Constraints	(none)

Table 33 — Elements of “Basic Observations::ObservationCharacteristics” (FeatureType)

Name	ObservationCharacteristics		
Definition	set of common characteristics used for describing one or more Observation		
Stereotype	FeatureType		
Inherits from	AbstractObservationCharacteristics		
Generalization of	ObservingCapability		
Abstract	False		
Associations	Name	Type	Definition
	ObservationCollection	ObservationCollection	(none)
	source:collection [0..*]	ObservationCollection	TODO
	target:memberCharacteristics [0..1]	ObservationCharacteristics	Information on ObservationCharacteristics of Observations contained within the ObservationCollection
Attributes	(none)		
Constraints	(none)		

Table 34 — Elements of “Basic Observations::ObservationCollection” (FeatureType)

Name	ObservationCollection		
Definition	A collection of Observations		
Stereotype	FeatureType		
Abstract	False		
Associations	Name	Type	Definition
	ObservationCharacteristics	ObservationCharacteristics	(none)
	source:collection [0..*]	ObservationCollection	TODO
	target:memberCharacteristics [0..1]	ObservationCharacteristics	Information on ObservationCharacteristics of Observations contained within the ObservationCollection
	ObservationCollection	ObservationCollection	(none)
	source:(self) [none]	ObservationCollection	(none)
	target:relatedCollection [0..*]	ObservationCollection	Another ObservationCollection the context ObservationCollection is related to
	Observation	Observation	(none)
	source:(self) [none]	ObservationCollection	(none)
	target:member [0..*]	Observation	an Observation that is part of this ObservationCollection
	Any	Any	(none)
	source:(self) [none]	ObservationCollection	(none)

Table 34 (continued)

	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	ObservationCollection	ObservationCollection	(none)
	source:(self) [none]	ObservationCollection	(none)
	target:relatedCollection [0..*]	ObservationCollection	Another ObservationCollection the context ObservationCollection is related to
Attributes	Name	Type	Definition
	collectionType [*]	AbstractObservationCollectionType	information on the type of the ObservationCollection
Constraints	(none)		

Table 35 — Elements of “Basic Observations::ObservationTypeByResultType” (CodeList)

Name	ObservationTypeByResultType
Definition	Observation classification scheme based on the type of its result property
Stereotype	CodeList
Inherits from	AbstractObservationType
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 36 — Elements of “Basic Observations::Observer” (FeatureType)

Name	Observer		
Definition	Concrete class for expressing a realization of Conceptual Observation Schema:Observer interface based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractObserver		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	link [*]	URI	Additional descriptive resources pertaining to a feature.
	location [0..1]	Geometry	Location information pertaining to a feature.
Constraints	(none)		

Table 37 — Elements of “Basic Observations::ObservingCapability” (FeatureType)

Name	ObservingCapability
Definition	Information on Observation (s) that could potentially be provided.
Stereotype	FeatureType

Table 37 (continued)

Inherits from	ObservationCharacteristics
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 38 — Elements of “Basic Observations::ObservingProcedure” (FeatureType)

Name	ObservingProcedure		
Definition	Concrete class for expressing a realization of Conceptual Observation Schema: ObservingProcedure interface based on General Feature Model as defined in ISO 19109		
Stereotype	FeatureType		
Inherits from	AbstractObservingProcedure		
Abstract	False		
Associations	(none)		
Attributes	<i>Name</i>	<i>Type</i>	<i>Definition</i>
	link [*]	URI	Additional descriptive resources pertaining to a feature.
Constraints	(none)		

Table 39 — Elements of “Basic Observations::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 40 — Elements of “Basic Observations::From ISO 19156 Observations and measurements — Conceptual observation schema” ()

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 41 — Elements of “Basic Observations::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface

Table 41 *(continued)*

Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 42 — Elements of “Basic Observations::From ISO 19156
Observations and measurements — Abstract observation core” ()**

Name	From ISO 19156 Observations and measurements — Abstract observation core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 43 — Elements of “Basic Observations::From ISO 19156
Observations and measurements — Conceptual observation schema” ()**

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 44 — Elements of “Basic Observations::From ISO 19156
Observations and measurements — Conceptual observation schema” ()**

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 45 — Elements of “Basic Observations::From ISO
19103 Conceptual schema language — Any type” ()**

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 46 — Elements of “Basic Observations::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 47 — Elements of “Basic Observations::From ISO 19156 Observations and measurements — Abstract observation core” ()

Name	From ISO 19156 Observations and measurements — Abstract observation core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 48 — Elements of “Basic Observations::From ISO 19156 Observations and measurements — Abstract observation core” ()

Name	From ISO 19156 Observations and measurements — Abstract observation core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 49 — Elements of “Basic Observations::From ISO 19156 Observations and measurements — Conceptual observation schema” ()

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

8 Basic Samples package

8.1 Basic Samples overview

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Figure 20 — Basic Samples - overview

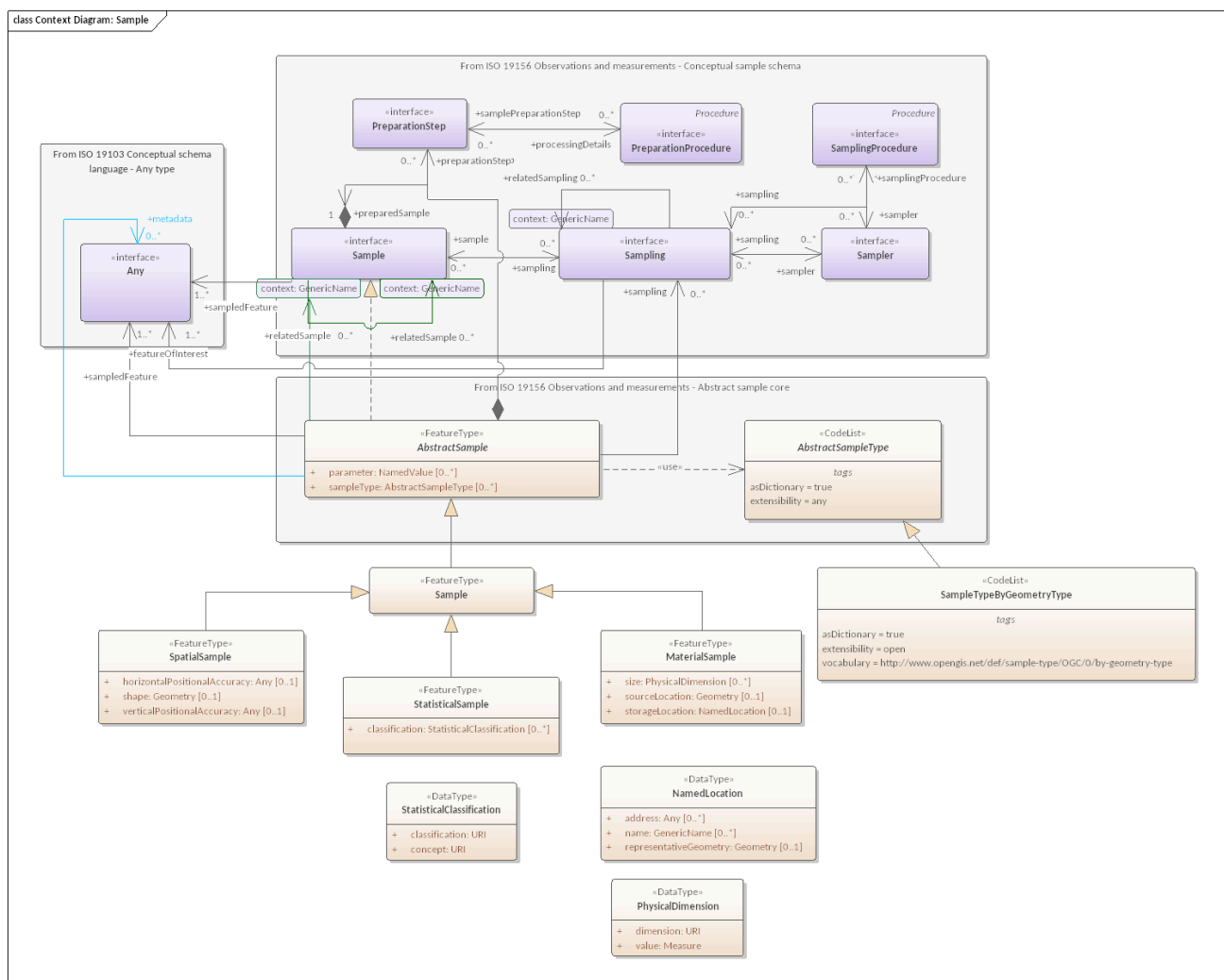


Figure 21 — Context Diagram: Sample

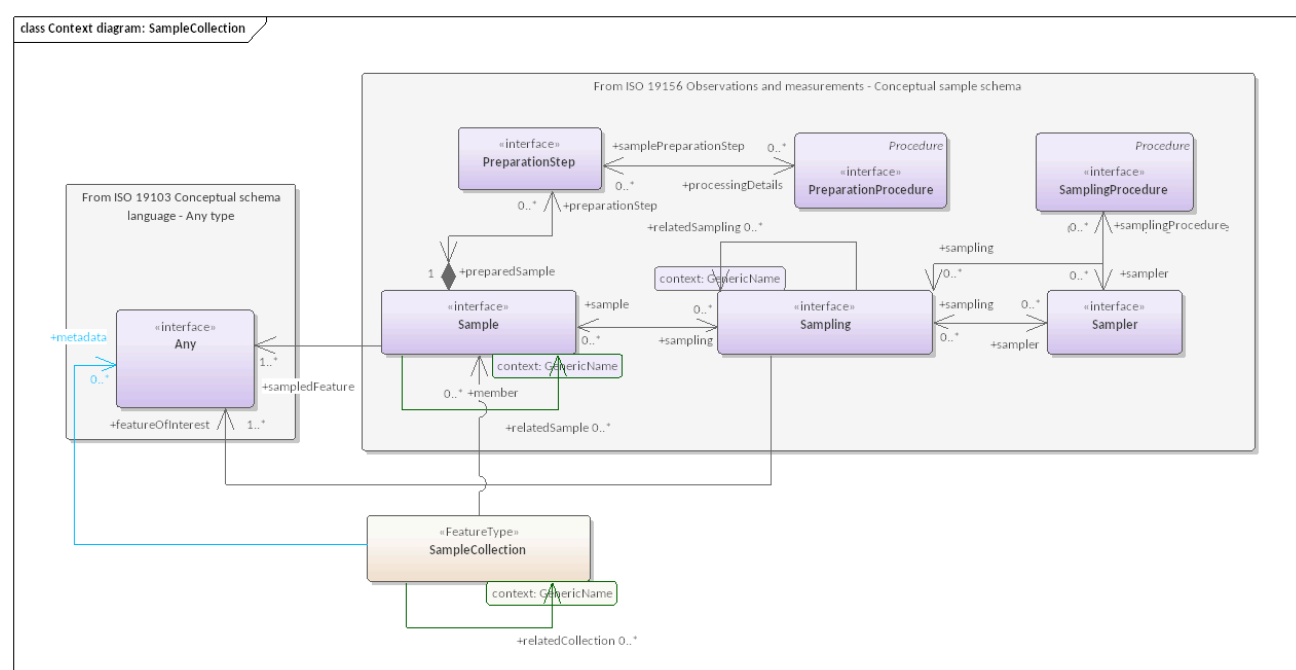


Figure 22 — Context diagram: SampleCollection

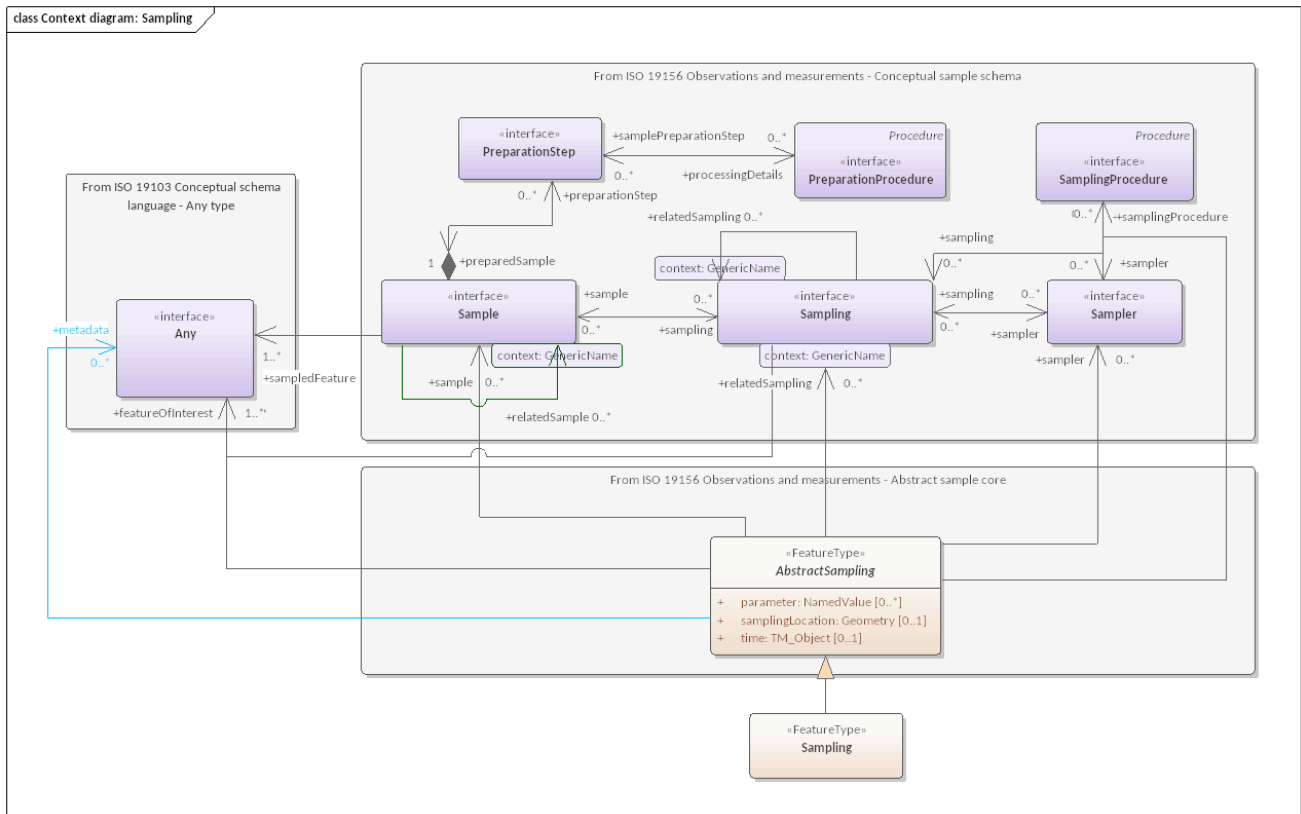


Figure 23 — Context diagram: Sampling

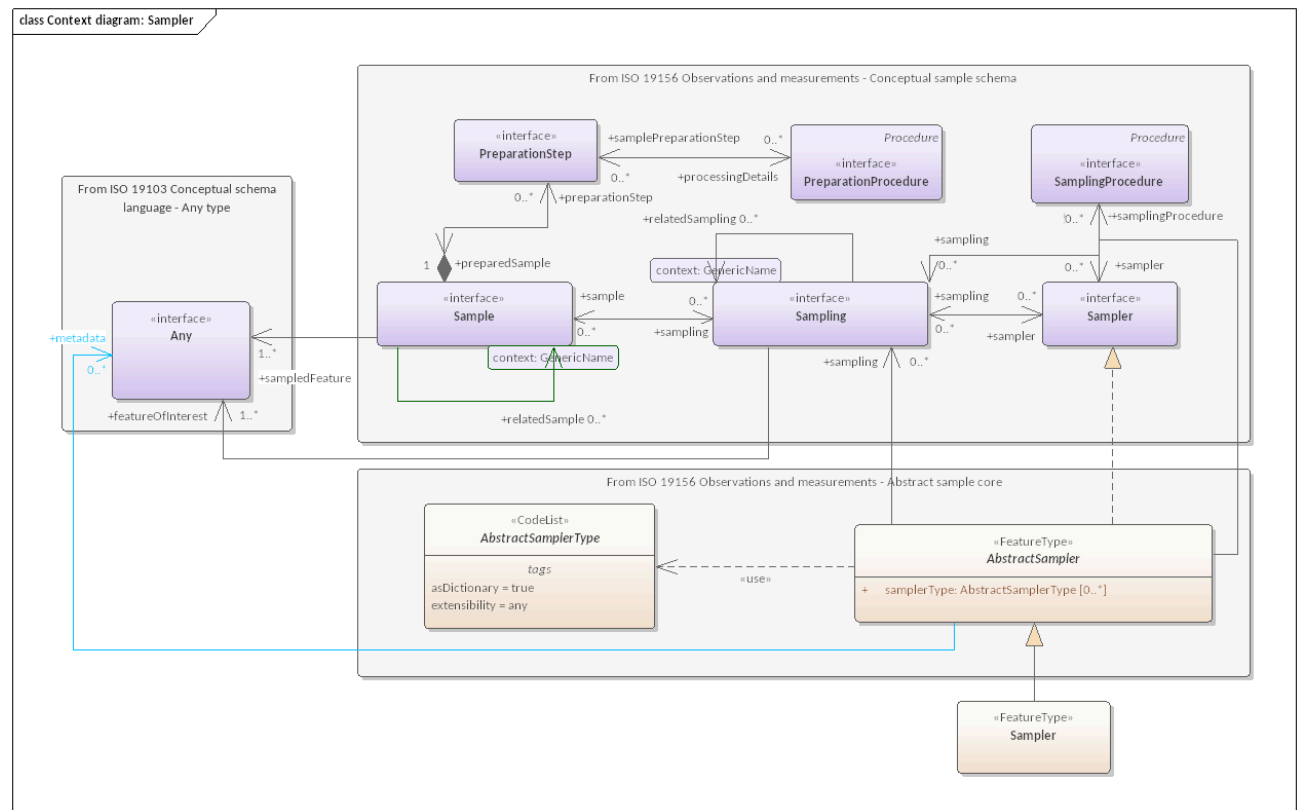


Figure 24 — Context diagram: Sampler

8.2 Defining tables

Table 50 — Elements of “Basic Samples::MaterialSample” (FeatureType)

Name	MaterialSample		
Definition	A MaterialSample is a physical, tangible Sample . NOTE 1 MaterialSamples that are curated and preserved are sometimes known as ‘specimens’. NOTE 2 MaterialSamples can be destroyed in connexion with the observation act. NOTE 3 A MaterialSample is a physical Sample of a FeatureOfInterest, obtained for Observation(s) normally carried out ex-situ, sometimes in a laboratory.		
Stereotype	FeatureType		
Inherits from	Sample		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	size [*]	PhysicalDimension	a physical extent of the MaterialSample
	sourceLocation [0..1]	Geometry	location from where the MaterialSample was obtained. TODO: Note text in spec
	storageLocation [0..1]	NamedLocation	location of a MaterialSample . NOTE The storageLocation may be a location such as a shelf in a warehouse or a drawer in a museum.
Constraints	(none)		

Table 51 — Elements of “Basic Samples::NamedLocation” (DataType)

Name	NamedLocation		
Definition	A location identified by its name, address, spatial geometry or a combination of any of these three		
Stereotype	DataType		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	address [*]	Any	An address used for identifying a NamedLocation
	name [*]	GenericName	A name used for identifying a NamedLocation .
	representativeGeometry [0..1]	Geometry	A geometry used for providing a representative spatial location of a NamedLocation
Constraints	(none)		

Table 52 — Elements of “Basic Samples::PhysicalDimension” (DataType)

Name	PhysicalDimension		
Definition	A dataType for the provision of various size quantities		
Stereotype	DataType		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	dimension [1]	URI	name of the PhysicalDimension about which a value is provided.

Table 52 (continued)

	value [1]	Measure	The value of the PhysicalDimension .
Constraints	(none)		

Table 53 — Elements of “Basic Samples::PreparationProcedure” (FeatureType)

Name	PreparationProcedure
Definition	Concrete class for expressing a realization of Conceptual Sample Schema:PreparationProcedure interface based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Inherits from	AbstractPreparationProcedure
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 54 — Elements of “Basic Samples::PreparationStep” (FeatureType)

Name	PreparationStep
Definition	Concrete class for expressing a realization of Conceptual Sample Schema:PreparationStep interface based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Inherits from	AbstractPreparationStep
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 55 — Elements of “Basic Samples::Sample” (FeatureType)

Name	Sample
Definition	Concrete class for expressing a realization of Conceptual Sample Schema:Sample interface based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Inherits from	AbstractSample
Generalization of	MaterialSample, SpatialSample, StatisticalSample
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 56 — Elements of “Basic Samples::SampleCollection” (FeatureType)

Name	SampleCollection
Definition	A collection of Samples
Stereotype	FeatureType
Abstract	False

Table 56 (continued)

Associations	Name	Type	Definition
	Any	Any	(none)
	source:(self) [none]	SampleCollection	(none)
	target:metadata [0..*]	Any	link to descriptive metadata as commonly understood by communities. NOTE Attention should be given not to reinvent semantic that is explicitly modelled in the OMS model.
	Sample	Sample	(none)
	source:(self) [none]	SampleCollection	(none)
	target:member [0..*]	Sample	A Sample that is part of this SampleCollection
	SampleCollection	SampleCollection	(none)
	source:(self) [none]	SampleCollection	(none)
	target:relatedCollection [0..*]	SampleCollection	Another SampleCollection the context SampleCollection is related to
	SampleCollection	SampleCollection	(none)
	source:(self) [none]	SampleCollection	(none)
	target:relatedCollection [0..*]	SampleCollection	Another SampleCollection the context SampleCollection is related to
Attributes	(none)		
Constraints	(none)		

Table 57 — Elements of “Basic Samples::Sampler” (FeatureType)

Name	Sampler
Definition	Concrete class for expressing a realization of Conceptual Sample Schema:Sampler interface based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Inherits from	AbstractSampler
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 58 — Elements of “Basic Samples::SampleTypeByGeometryType” (CodeList)

Name	SampleTypeByGeometryType
Definition	SpatialSample classification scheme based on the geometry type of its shape attribute
Stereotype	CodeList
Inherits from	AbstractSampleType
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 59 — Elements of “Basic Samples::Sampling” (FeatureType)

Name	Sampling
Definition	Concrete class for expressing a realization of Conceptual Sample Schema:Sampling interface based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Inherits from	AbstractSampling
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 60 — Elements of “Basic Samples::SamplingProcedure” (FeatureType)

Name	SamplingProcedure
Definition	Concrete class for expressing a realization of Conceptual Sample Schema:SamplingProcedure interface based on General Feature Model as defined in ISO 19109
Stereotype	FeatureType
Inherits from	AbstractSamplingProcedure
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 61 — Elements of “Basic Samples::SpatialSample” (FeatureType)

Name	SpatialSample		
Definition	a geospatial Sample. NOTE When observations are made to estimate properties of a geospatial feature, in particular where the value of a property varies within the scope of the feature, a SpatialSample is used. Depending on accessibility and on the nature of the expected property variation, the SpatialSample may be extensive in one, two or three spatial dimensions. EXAMPLE — Typically an Observation ?site? or ‘station’ connotes the ‘world in the vicinity of the site (or station)’, so the observed properties relate to the physical medium at the station, and not to any physical artifact such as a mooring, buoy, benchmark, monument, well, etc. — Some common names for SpatialSample used in various application domains include Borehole, Flightline, Interval, Lidar Cloud, Map Horizon, Microscope Slide, Mine Level, Mine, Observation Well, Profile, Pulp, Quadrat, Scene, Section, ShipsTrack, Spot, Station, Swath, Trajectory, Traverse, etc.		
Stereotype	FeatureType		
Inherits from	Sample		
Abstract	False		
Associations	(none)		
Attributes	<i>Name</i>	<i>Type</i>	<i>Definition</i>
	horizontalPositionalAccuracy [0..1]	Any	The positional accuracy of the horizontal component of the Geometry of the SpatialSample .
	shape [0..1]	Geometry	Geometry of the SpatialSample . NOTE The shape of the SpatialSample is the context

Table 61 (continued)

			for domain decomposition. EXAMPLE Logs of different properties along a well or borehole might use different intervals, and sub-samples might be either spatially instantaneous, or averaged in some way over an interval. The position of the samples can be conveniently described in terms of offsets in a linear coordinate reference system that is defined by the shape of the well axis.
	verticalPositionalAccuracy [0..1]	Any	The positional accuracy of the vertical component of the Geometry of the SpatialSample
Constraints	(none)		

Table 62 — Elements of “Basic Samples::StatisticalClassification” (DataType)

Name	StatisticalClassification		
Definition	a dataType for the provision of information on statistical classifications.		
Stereotype	DataType		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	classification [1]	URI	The explicit classification class pertaining to the classification concept described by the StatisticalClassification . EXAMPLE The classification for a statistical classification could be: — Age Brackets: [0-10], [10-20] — Genders: Male, Female, Other — Color: Red, Green, Blue
	concept [1]	URI	The concept by which a StatisticalClassification is to be performed. EXAMPLE The concept for a statistical classification could be age, gender, color, size etc.
Constraints	(none)		

Table 63 — Elements of “Basic Samples::StatisticalSample” (FeatureType)

Name	StatisticalSample		
Definition	a statistical subset of a feature-of-interest, defined for the purpose of creating Observation(s) . NOTE 1 StatisticalSamples usually apply to populations or other sets, of which certain subset may be of specific interest. EXAMPLE The male or female subset of a population.		
Stereotype	FeatureType		
Inherits from	Sample		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	classification [*]	StatisticalClassification	a criterion by which the subset was defined

Table 63 (*continued*)

Constraints	(none)
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**Table 64 — Elements of “Basic Samples::From ISO 19156
Observations and measurements — Abstract sample core” ()**

Name	From ISO 19156 Observations and measurements — Abstract sample core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 65 — Elements of “Basic Samples::From ISO 19156
Observations and measurements — Conceptual sample schema” ()**

Name	From ISO 19156 Observations and measurements — Conceptual sample schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 66 — Elements of “Basic Samples::From ISO 19156
Observations and measurements — Abstract sample core” ()**

Name	From ISO 19156 Observations and measurements — Abstract sample core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 67 — Elements of “Basic Samples::From ISO 19156
Observations and measurements — Abstract sample core” ()**

Name	From ISO 19156 Observations and measurements — Abstract sample core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

**Table 68 — Elements of “Basic Samples::From ISO
19103 Conceptual schema language — Any type” ()**

Name	From ISO 19103 Conceptual schema language — Any type
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Table 68 (continued)

Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

9 Conceptual Observation schema package

9.1 Conceptual Observation schema overview

... ..

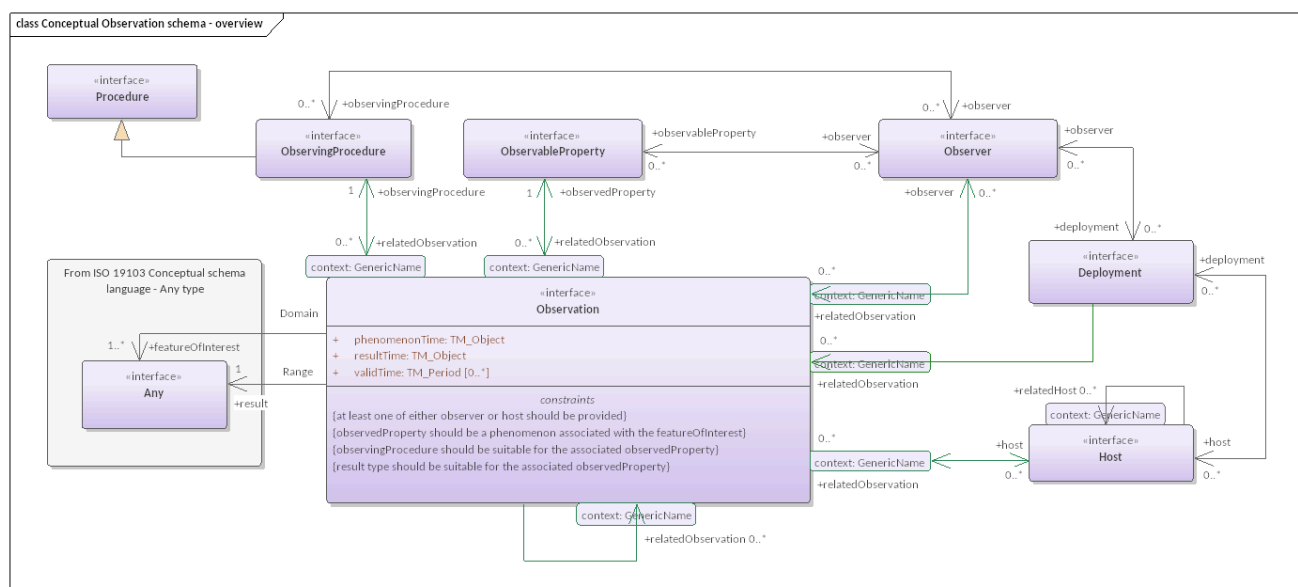


Figure 25 — Conceptual Observation schema - overview

9.2 Defining tables

Table 69 — Elements of “Conceptual Observation schema::Deployment” ()

Name	Deployment		
Definition	information on the assignment of an Observer to a Host. NOTE Examples of deployment are — information regarding a sensor being attached to a pole — the monitoring facilities pertaining to an environmental monitoring network — the description of a ship cruise linking a research vessel with a marine network — the participation of a citizen in a citizen-science project involving crowd sensing.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Observation	Observation	(none)
	source:(self) [none]	Deployment	(none)

Table 69 (continued)

target: relatedObservation [0..*]	Observation	An Observation related to the Deployment
Observer	Observer	(none)
source:observer [0..*]	Observer	Observer associated with this Deployment
target:deployment [0..*]	Deployment	Deployment to which this Observer is either physically or organizationally attached.
AbstractHost	AbstractHost	(none)
source:(self) [none]	AbstractHost	(none)
target:deployment [0..*]	Deployment	Deployment at this Host
AbstractObserver	AbstractObserver	(none)
source:(self) [none]	AbstractObserver	(none)
target:deployment [0..*]	Deployment	Deployment to which this Observer is either physically or organizationally attached.
Host	Host	(none)
source:host [0..*]	Host	Host to which this Deployment pertains
target:deployment [0..*]	Deployment	Deployment at this Host
Attributes	(none)	
Constraints	(none)	

Table 70 — Elements of “Conceptual Observation schema::Host” ()

Name	Host		
Definition	a grouping of Observers for a specific reason.NOTES: — In many use cases, the Host is the environmental monitoring facility — The Host can be a platform that hosts a set of sensors — An alternative usage could pertain to a biodiversity survey campaign; in this scenario, the team performing the survey would be modelled as observers whereas the entire survey campaign can be represented as a Host.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Host	Host	(none)
	source:(self) [none]	Host	(none)
	target:relatedHost [0..*]	Host	Another Host the context Host is related to
	Deployment	Deployment	(none)
	source:host [0..*]	Host	Host to which this Deployment pertains
	target:deployment [0..*]	Deployment	Deployment at this Host
	Observation	Observation	(none)
	source:host [0..*]	Host	Host that is involved in the creation of this Observation

Table 70 (continued)

target:relatedObservation [0..*]	Observation	An Observation related to the Host
AbstractDeployment	AbstractDeployment	(none)
source:(self) [none]	AbstractDeployment	(none)
target:host [0..*]	Host	Host to which this Deployment pertains
AbstractObservationCharacteristics	AbstractObservationCharacteristics	(none)
source:(self) [none]	AbstractObservationCharacteristics	(none)
target:host [0..*]	Host	Host that is involved in the creation of the Observation or a collection of Observations described
Host	Host	(none)
source:(self) [none]	Host	(none)
target:relatedHost [0..*]	Host	Another Host the context Host is related to
AbstractHost	AbstractHost	(none)
source:(self) [none]	AbstractHost	(none)
target:relatedHost [0..*]	Host	Another Host the context Host is related to
Attributes	(none)	
Constraints	(none)	

Table 71 — Elements of “Conceptual Observation schema::ObservableProperty” ()

Name	ObservableProperty		
Definition	An observable quality (property, characteristic) of the feature of interest that can be observed. EXAMPLE 1 The height of a tree, the depth of a water body, or the temperature of a surface are examples of observable properties, while the value of a classic car is not (directly) observable but asserted. EXAMPLE 2 Groundwater Level On a groundwater well we — monitor Groundwater Level (1 observable property) — with an automated probe (that remains in the ground all year, constituting 1 procedure). Then we have physical campaigns where we revisit the groundwater well and — measure the Groundwater Level (still the same observable property as above) — but with a manual probe, this is a different procedure. This allows to check whether the probe needs recalibration.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	AbstractObserver	AbstractObserver	(none)
	source:(self) [none]	AbstractObserver	(none)

Table 71 (continued)

target:observableProperty [0..*]	ObservableProperty	ObservableProperty that this Observer can observe
Observation	Observation	(none)
source:relatedObservation [0..*]	Observation	An Observation related to the ObservableProperty
target:observedProperty [1]	ObservableProperty	ObservableProperty that is the subject of the Observation.
Observer	Observer	(none)
source:observer [0..*]	Observer	Observer capable of observing this ObservableProperty
target:observableProperty [0..*]	ObservableProperty	ObservableProperty that this Observer can observe
AbstractObservationCharacteristics	AbstractObservationCharacteristics	(none)
source:(self) [none]	AbstractObservationCharacteristics	(none)
target:observedProperty [0..*]	ObservableProperty	ObservableProperty that is the subject of the Observation or a collection of Observations described
Attributes	(none)	
Constraints	(none)	

Table 72 — Elements of “Conceptual Observation schema::Observation” ()

Name	Observation		
Definition	an act carried out by an observer to determine the value of an observable property of an object (feature of interest) by using a procedure; the value is provided as the result. NOTE It is important to note that the terms ?observation?, ?interpretation?, ?forecast?, ?simulation? do correspond to this definition. This aspect is further clarified in Clause 7.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Observer	Observer	(none)
	source:relatedObservation [0..*]	Observation	An Observation related to the Observer
	target:observer [0..*]	Observer	Observer that is involved in the creation of this Observation
	Domain	Any	(none)
	source:(self) [none]	Observation	(none)
	target:featureOfInterest [1..*]	Any	subject of the Observation.
	Range	Any	(none)
	source:(self) [none]	Observation	(none)
	target:result [1]	Any	result of the Observation
	ObservableProperty	ObservableProperty	(none)

Table 72 (continued)

source:relatedObservation [0..*]	Observation	An Observation related to the ObservableProperty
target:observedProperty [1]	ObservableProperty	ObservableProperty that is the subject of the Observation.
ObservingProcedure	ObservingProcedure	(none)
source:relatedObservation [0..*]	Observation	Observation related to the ObservingProcedure
target:observingProcedure [1]	ObservingProcedure	ObservingProcedure used by the Observation to determine the value of the ObservableProperty provided by the result.
Observation	Observation	(none)
source:(self) [none]	Observation	(none)
target:relatedObservation [0..*]	Observation	Another Observation the context Observation is related to
AbstractSample	AbstractSample	(none)
source:(self) [none]	AbstractSample	(none)
target:relatedObservation [0..*]	Observation	An Observation related to the sample
Sample	Sample	(none)
source:(self) [none]	Sample	(none)
target:relatedObservation [0..*]	Observation	An Observation related to the Sample
AbstractObservation	AbstractObservation	(none)
source:(self) [none]	AbstractObservation	(none)
target:relatedObservation [0..*]	Observation	Another Observation related to the context Observation
Deployment	Deployment	(none)
source:(self) [none]	Deployment	(none)
target:relatedObservation [0..*]	Observation	An Observation related to the Deployment
AbstractDeployment	AbstractDeployment	(none)
source:(self) [none]	AbstractDeployment	(none)
target:relatedObservation [0..*]	Observation	An Observation related to the deployment
AbstractObserver	AbstractObserver	(none)
source:(self) [none]	AbstractObserver	(none)
target:relatedObservation [0..*]	Observation	An Observation related to the observer
AbstractObservingProcedure	AbstractObservingProcedure	(none)
source:(self) [none]	AbstractObservingProcedure	(none)
target:relatedObservation [0..*]	Observation	An Observation related to the observing procedure
Host	Host	(none)
source:host [0..*]	Host	Host that is involved in the creation of this Observation
target:relatedObservation [0..*]	Observation	An Observation related to the Host

Table 72 (continued)

	GenericDomainFeature	GenericDomainFeature	(none)
	source:(self) [none]	GenericDomainFeature	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the domain feature
	AbstractHost	AbstractHost	(none)
	source:(self) [none]	AbstractHost	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the host
	ObservationCollection	ObservationCollection	(none)
	source:(self) [none]	ObservationCollection	(none)
	target:member [0..*]	Observation	an Observation that is part of this ObservationCollection
	AbstractObservableProperty	AbstractObservableProperty	(none)
	source:(self) [none]	AbstractObservableProperty	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the observable property
	Observation	Observation	(none)
	source:(self) [none]	Observation	(none)
	target:relatedObservation [0..*]	Observation	Another Observation the context Observation is related to
Attributes	Name	Type	Definition
	phenomenonTime [1]	TM_Object	time that the Result applies to the characteristic of the FeatureOfInterest being observed. NOTE 1 The phenomenonTime is often the time the Sample has been taken. NOTE 2 This is often the time of interaction by a sampling Procedure or observation Procedure with a real-world feature. NOTE 3 If the result is the average of multiple samples taken at different times, then the phenomenonTime is the time interval over which these measurements were taken.
	resultTime [1]	TM_Object	instant of time when the result of the Observation became available. EXAMPLE 1 The resultTime typically corresponds to when the Procedure associated with the Observation was completed. For some observations this is identical to the phenomenonTime. However, there are important cases where they differ. EXAMPLE 2 Where a measurement is made on a specimen in a laboratory, the phenomenonTime is the time the specimen was retrieved from its host, while the resultTime is the time the laboratory procedure was applied.

Table 72 (continued)

			EXAMPLE 3 The resultTime also supports disambiguation of repeat measurements made of the same property of a feature using the same procedure. EXAMPLE 4 Where sensor observation results are post-processed, the resultTime is the post-processing time, while the phenomenonTime is the time of initial interaction with the world. EXAMPLE 5 Simulations may be used to estimate the values for phenomena in the future or past. The phenomenonTime is the time that the result applies to, while the resultTime is the time that the simulation was executed.
	validTime [*]	TM_Period	The time interval during which the result is assumed to be applicable for use. NOTE This attribute is commonly required in forecasting applications.
Constraints	(none)		

Table 73 — Elements of “Conceptual Observation schema::Observer” ()

Name	Observer		
Definition	identifiable entity that can generate Observations pertaining to an observableProperty by implementing a Procedure. NOTES — Different Observers can follow the same (reusable) observing Procedure for the creation of different Observations. — The Observer is the entity instance, not the entity type. Pertaining to sensors, the Observer would reference the explicit sensor, while the Procedure would reference the methodology utilized by that sensor type. — An Observer is closely linked with an observableProperty that it generates results for. — An Observer can be hosted by one or more Host. — The Observer is an instance of a sensor, instrument, implementation of an algorithm or a being such as a person. An Observer responds to a stimulus, e.g., a change in the environment, or input data composed from the Results of prior Observations, and generates a Result. EXAMPLE Accelerometers, gyroscopes, barometers, magnetometers, and so forth are Observers that are typically mounted on a modern smartphone (which acts as Host). Other examples of Sensors include the human eyes.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Deployment	Deployment	(none)
	source:observer [0..*]	Observer	Observer associated with this Deployment
	target:deployment [0..*]	Deployment	Deployment to which this Observer is either physically

Table 73 (continued)

		or organizationally attached.
ObservableProperty	ObservableProperty	(none)
source:observer [0..*]	Observer	Observer capable of observing this ObservableProperty
target:observableProperty [0..*]	ObservableProperty	ObservableProperty that this Observer can observe
ObservingProcedure	ObservingProcedure	(none)
source:observer [0..*]	Observer	Observer capable of performing this ObservingProcedure.
target:observingProcedure [0..*]	ObservingProcedure	An ObservingProcedure that this Observer can perform
AbstractObservableProperty	AbstractObservableProperty	(none)
source:(self) [none]	AbstractObservableProperty	(none)
target:observer [0..*]	Observer	Observer capable of observing this ObservableProperty
Observation	Observation	(none)
source:relatedObservation [0..*]	Observation	An Observation related to the Observer
target:observer [0..*]	Observer	Observer that is involved in the creation of this Observation
AbstractObservingProcedure	AbstractObservingProcedure	(none)
source:(self) [none]	AbstractObservingProcedure	(none)
target:observer [0..*]	Observer	Observer capable of performing this ObservingProcedure.
AbstractObservationCharacteristics	AbstractObservationCharacteristics	(none)
source:(self) [none]	AbstractObservationCharacteristics	(none)
target:observer [0..*]	Observer	Observer that is involved in the creation of Observation or a collection of Observations described
AbstractDeployment	AbstractDeployment	(none)
source:(self) [none]	AbstractDeployment	(none)
target:observer [0..*]	Observer	Observer associated with this Deployment
Attributes	(none)	

Table 73 (continued)

Constraints	(none)
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Table 74 — Elements of “Conceptual Observation schema::ObservingProcedure” ()

Name	ObservingProcedure		
Definition	description of steps performed in order to determine the value of an observableProperty by an Observer. NOTES: — Depending on the complexity of the use case, the procedure will be more or less explicitly described. Especially pertaining to historical data, there may be very little or no information available — this information should also be provided. — The recipe that the observer (cook) follows to generate the observation. — The procedure is often referred to as the method. — Different observers can follow the same (reusable) procedure for the creation of different observations. — The procedure is a workflow, protocol, plan, algorithm, or computational method specifying how to make an observation. — the observing procedure cannot describe a sensor instance, but it can describe the sensor type. NOTE The term process that was used in ISO 19156:2011 has been purposely dropped in this version to avoid unnecessary confusion between the terms procedure and process. EXAMPLE An instance of Procedure is a description of the process utilized by an observer, this could be a chemical analysis method, a protocol for measuring an object, but could also be a checklist utilized by a human observer during a biodiversity campaign. It could further describe the algorithms behind simulators or models used to generate a result from other inputs.		
Stereotype	interface		
Inherits from	Procedure		
Abstract	True		
Associations	Name	Type	Definition
	AbstractObserver	AbstractObserver	(none)
	source:(self) [none]	AbstractObserver	(none)
	target:observingProcedure [0..*]	ObservingProcedure	An ObservingProcedure that this Observer can perform
	AbstractObservationCharacteristics	AbstractObservationCharacteristics	(none)
	source:(self) [none]	AbstractObservationCharacteristics	(none)
	target:observingProcedure [0..*]	ObservingProcedure	ObservingProcedure used by the Observation or a collection of Observations described to determine the value of the ObservableProperty provided by the result.
	Observer	Observer	(none)

Table 74 (continued)

	source:observer [0..*]	Observer	Observer capable of performing this ObservingProcedure.
	target:observingProcedure [0..*]	ObservingProcedure	An ObservingProcedure that this Observer can perform
	Observation	Observation	(none)
	source:relatedObservation [0..*]	Observation	Observation related to the ObservingProcedure
	target:observingProcedure [1]	ObservingProcedure	ObservingProcedure used by the Observation to determine the value of the ObservableProperty provided by the result.
Attributes	(none)		
Constraints	(none)		

Table 75 — Elements of “Conceptual Observation schema::Procedure” ()

Name	Procedure
Definition	description of steps performed. NOTE Procedure is an abstract concept that is then further specialized in the various procedure types defined in this document. All share the commonality of describing a defined series of steps to a specific purpose. NOTE The term <i>process</i> that was used in ISO 19156:2011 has been purposely dropped in this version to avoid unnecessary confusion between the terms procedure and process.
Stereotype	interface
Generalization of	ObservingProcedure, SamplingProcedure, PreparationProcedure
Abstract	True
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 76 — Elements of “Conceptual Observation schema::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

10 Conceptual Sample schema package

10.1 Conceptual Sample schema overview

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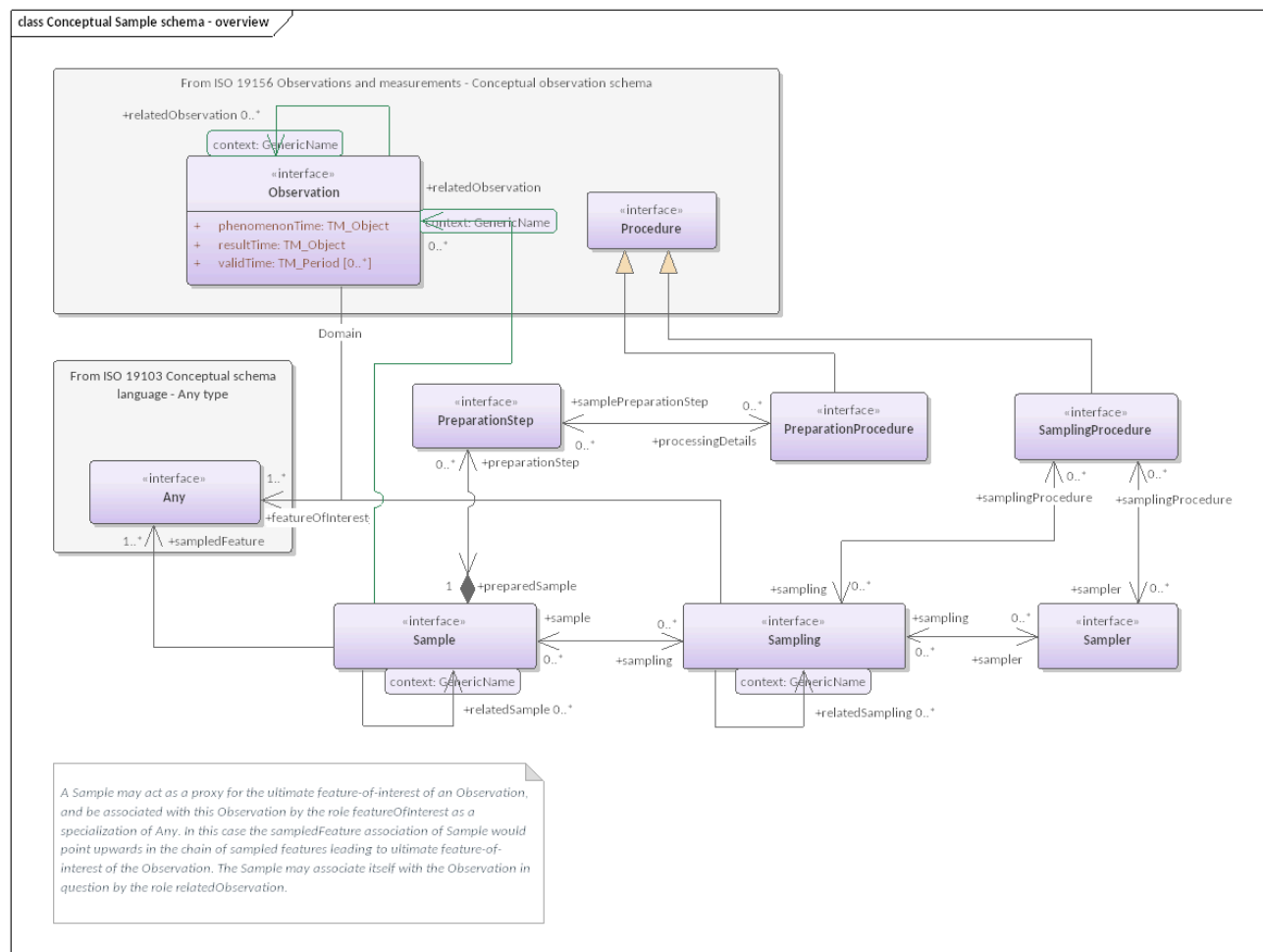


Figure 26 — Conceptual Sample schema - overview

10.2 Defining tables

Table 77 — Elements of “Conceptual Sample schema::PreparationProcedure” ()

Name	PreparationProcedure		
Definition	description of preparation steps performed on a Sample.		
Stereotype	interface		
Inherits from	Procedure		
Abstract	True		
Associations	<i>Name</i>	<i>Type</i>	<i>Definition</i>
	AbstractPreparationStep	AbstractPreparationStep	(none)
	source:(self) [none]	AbstractPreparationStep	(none)
	target:processingDetails [0..*]	PreparationProcedure	PreparationProcedure step performed on the Sample the PreparationStep pertains to
	PreparationStep	PreparationStep	(none)

Table 77 (continued)

	source: samplePreparationStep [0..*]	PreparationStep	provides information on the PreparationSteps where this procedure has been used
	target:processingDetails [0..*]	PreparationProcedure	PreparationProcedure step performed on the Sample the PreparationStep pertains to
Attributes	(none)		
Constraints	(none)		

Table 78 — Elements of “Conceptual Sample schema::PreparationStep” ()

Name	PreparationStep		
Definition	an individual step pertaining to a PreparationProcedure.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	PreparationProcedure	PreparationProcedure	(none)
	source:samplePreparationStep [0..*]	PreparationStep	provides information on the PreparationSteps where this procedure has been used
	target:processingDetails [0..*]	PreparationProcedure	PreparationProcedure step performed on the Sample the PreparationStep pertains to
	AbstractSample	AbstractSample	(none)
	source:(self) [none]	AbstractSample	(none)
	target:preparationStep [0..*]	PreparationStep	PreparationStep(s) applied to prepare the Sample
	Sample	Sample	(none)
	source:preparedSample [1]	Sample	Sample on which the PreparationProcedure is performed
	target:preparationStep [0..*]	PreparationStep	PreparationStep(s) applied to prepare the Sample
	AbstractPreparationProcedure	AbstractPreparationProcedure	(none)
	source:(self) [none]	AbstractPreparationProcedure	(none)
	target:samplePreparationStep [0..*]	PreparationStep	provides information on the PreparationSteps where this procedure has been used
Attributes	(none)		
Constraints	(none)		

Table 79 — Elements of “Conceptual Sample schema::Sample” ()

Name	Sample
Definition	an object that is representative of a concept, real-world object or phenomenon. NOTE — The way the sample is taken is typically guided by a sampling strategy. Samples are often artefacts of an observational strategy, and have often no significant function outside of their role in the observation process (although ?specimen preservation could be considered a specific activity per se?).

Table 79 (continued)

	— The physical characteristics of the features themselves are of little interest, except perhaps to the manager of a sampling campaign. — Typically, the Sample is a Feature which is intended to be representative of a FeatureOfInterest on which Observations may be made. As such, it may carry a characteristic pertaining to the observedProperty being evaluated by the Observation. EXAMPLE 1 A profile typically samples a water- or atmospheric-column; a well samples the water in an aquifer; a tissue specimen samples a part of an organism. EXAMPLE 2 A statistical sample is often designed to be characteristic of an entire population, so that Observations can be made regarding the sample that provide a good estimate of the properties of the population.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Sample	Sample	(none)
	source:(self) [none]	Sample	(none)
	target:relatedSample [0..*]	Sample	Another Sample the context Sample is related to
	PreparationStep	PreparationStep	(none)
	source:preparedSample [1]	Sample	Sample on which the PreparationProcedure is performed
	target:preparationStep [0..*]	PreparationStep	PreparationStep(s) applied to prepare the Sample
	Observation	Observation	(none)
	source:(self) [none]	Sample	(none)
	target:relatedObservation [0..*]	Observation	An Observation related to the Sample
	Sampling	Sampling	(none)
	source:sample [0..*]	Sample	Sample generated by the Sampling
	target:sampling [0..*]	Sampling	Sampling the Sample is the result of
	Any	Any	(none)
	source:(self) [none]	Sample	(none)
	target:sampledFeature [1..*]	Any	the feature the Sample is intended to be representative of. NOTE The sampled feature is usually a real-world feature from an application domain.
	AbstractPreparationStep	AbstractPreparationStep	(none)
	source:(self) [none]	AbstractPreparationStep	(none)
	target:preparedSample [1]	Sample	Sample on which the PreparationProcedure is performed
	Sample	Sample	(none)
	source:(self) [none]	Sample	(none)
	target:relatedSample [0..*]	Sample	Another Sample the context Sample is related to
	SampleCollection	SampleCollection	(none)
	source:(self) [none]	SampleCollection	(none)

Table 79 (continued)

	target:member [0..*]	Sample	A Sample that is part of this SampleCollection
	AbstractSampling	AbstractSampling	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:sample [0..*]	Sample	Sample generated by the Sampling
	AbstractSample	AbstractSample	(none)
	source:(self) [none]	AbstractSample	(none)
	target:relatedSample [0..*]	Sample	Another Sample the context Sample is related to
Attributes	(none)		
Constraints	(none)		

Table 80 — Elements of “Conceptual Sample schema::Sampler” ()

Name	Sampler		
Definition	a device or entity (including humans) that is used by, or implements, a SamplingProcedure to create or transform one or more Sample(s). EXAMPLES — A ball mill, diamond drill, hammer, — hypodermic syringe and needle, — image sensor, a soil auger, — a human being. NOTE All the examples above can act as sampling devices (i.e., be Samplers). However, sometimes the distinction between the Sampler and the Sensor is not evident, as they are packaged as a unit. A Sampler need not be a physical device.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	SamplingProcedure	SamplingProcedure	(none)
	source:sampler [0..*]	Sampler	provides information on the Sampler that implements this procedure
	target:samplingProcedure [0..*]	SamplingProcedure	Procedure implemented by the Sampler
	AbstractSampling	AbstractSampling	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:sampler [0..*]	Sampler	Sampler that performed the Sampling
	Sampling	Sampling	(none)
	source:sampling [0..*]	Sampling	Sampling act performed by the Sampler
	target:sampler [0..*]	Sampler	Sampler that performed the Sampling
	AbstractSamplingProcedure	AbstractSamplingProcedure	(none)
	source:(self) [none]	AbstractSamplingProcedure	(none)
	target:sampler [0..*]	Sampler	provides information on the Sampler that

Table 80 (continued)

		implements this procedure
Attributes	(none)	
Constraints	(none)	

Table 81 — Elements of “Conceptual Sample schema::Sampling” ()

Name	Sampling		
Definition	an act applying a SamplingProcedure to create or transform one or more Sample(s). EXAMPLES — Crushing a rock sample in a ball mill, — Digging a pit through a soil sequence, — Dividing a field site into quadrants, — Drawing blood from a patient, — Extracting water from an observation well, — Extracting a sample from a defined environmental monitoring station, — Registering an image of the landscape, — Sieving a powder to separate the subset finer than 100-mesh, — Selecting a subset of a population, — Splitting a piece of drill-core to create two new samples, — Taking a diamond-drill core from a rock outcrop.		
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Sampling	Sampling	(none)
	source:(self) [none]	Sampling	(none)
	target:relatedSampling [0..*]	Sampling	Another Sampling the context Sampling is related to
	Any	Any	(none)
	source:(self) [none]	Sampling	(none)
	target:featureOfInterest [1..*]	Any	The concept, real-world object or phenomenon (feature-of-interest) the Sample(s) of the Sampling represent
	Sampler	Sampler	(none)
	source:sampling [0..*]	Sampling	Sampling act performed by the Sampler
	target:sampler [0..*]	Sampler	Sampler that performed the Sampling
	SamplingProcedure	SamplingProcedure	(none)
	source:sampling [0..*]	Sampling	provides information on the Sampling where this procedure has been used
	target:samplingProcedure [0..*]	SamplingProcedure	SamplingProcedure used by the Sampling
	Sampling	Sampling	(none)
	source:(self) [none]	Sampling	(none)

Table 81 (continued)

target:relatedSampling [0..*]	Sampling	Another Sampling the context Sampling is related to
AbstractSample	AbstractSample	(none)
source:(self) [none]	AbstractSample	(none)
target:sampling [0..*]	Sampling	Sampling the Sample is the result of
AbstractSamplingProcedure	AbstractSamplingProcedure	(none)
source:(self) [none]	AbstractSamplingProcedure	(none)
target:sampling [0..*]	Sampling	provides information on the Sampling where this procedure has been used
AbstractSampling	AbstractSampling	(none)
source:(self) [none]	AbstractSampling	(none)
target:relatedSampling [0..*]	Sampling	Another Sampling the context Sampling is related to
Sample	Sample	(none)
source:sample [0..*]	Sample	Sample generated by the Sampling
target:sampling [0..*]	Sampling	Sampling the Sample is the result of
AbstractSampler	AbstractSampler	(none)
source:(self) [none]	AbstractSampler	(none)
target:sampling [0..*]	Sampling	Sampling act performed by the Sampler
Attributes	(none)	
Constraints	(none)	

Table 82 — Elements of “Conceptual Sample schema::SamplingProcedure” ()

Name	SamplingProcedure		
Definition	description of steps performed by a Sampler in order to extract a Sample from its sampledFeature in the frame of a Sampling.		
Stereotype	interface		
Inherits from	Procedure		
Abstract	True		
Associations	Name	Type	Definition
	AbstractSampling	AbstractSampling	(none)
	source:(self) [none]	AbstractSampling	(none)
	target:samplingProcedure [0..*]	SamplingProcedure	SamplingProcedure used by the Sampling
	AbstractSampler	AbstractSampler	(none)
	source:(self) [none]	AbstractSampler	(none)
	target:samplingProcedure [0..*]	SamplingProcedure	Procedure implemented by the Sampler
	Sampling	Sampling	(none)
	source:sampling [0..*]	Sampling	provides information on the Sampling where this procedure has been used
	target:samplingProcedure [0..*]	SamplingProcedure	SamplingProcedure used by the Sampling
	Sampler	Sampler	(none)

Table 82 (continued)

	source:sampler [0..*]	Sampler	provides information on the Sampler that implements this procedure
	target: samplingProcedure [0..*]	SamplingProcedure	Procedure implemented by the Sampler
Attributes	(none)		
Constraints	(none)		

Table 83 — Elements of “Conceptual Sample schema::From ISO 19156 Observations and measurements — Conceptual observation schema” ()

Name	From ISO 19156 Observations and measurements — Conceptual observation schema
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 84 — Elements of “Conceptual Sample schema::From ISO 19103 Conceptual schema language — Any type” ()

Name	From ISO 19103 Conceptual schema language — Any type
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

11 Examples package

11.1 Examples overview

... ..

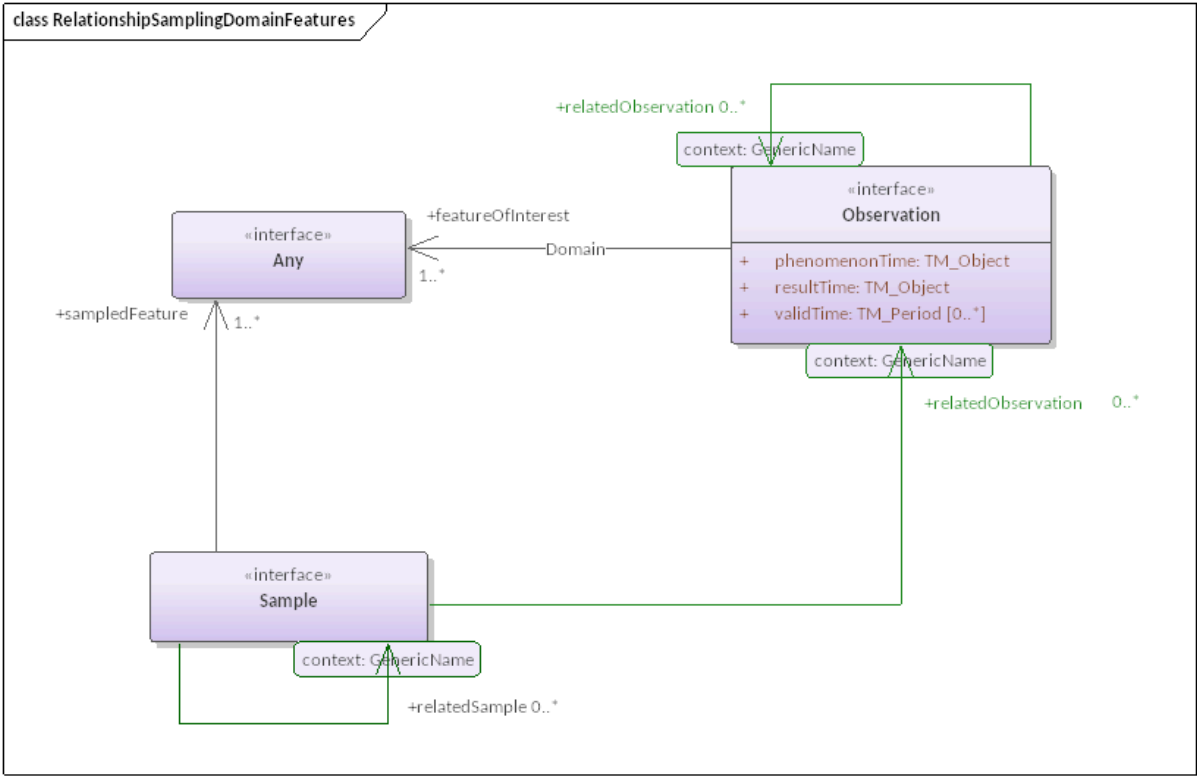


Figure 27 — RelationshipSamplingDomainFeatures

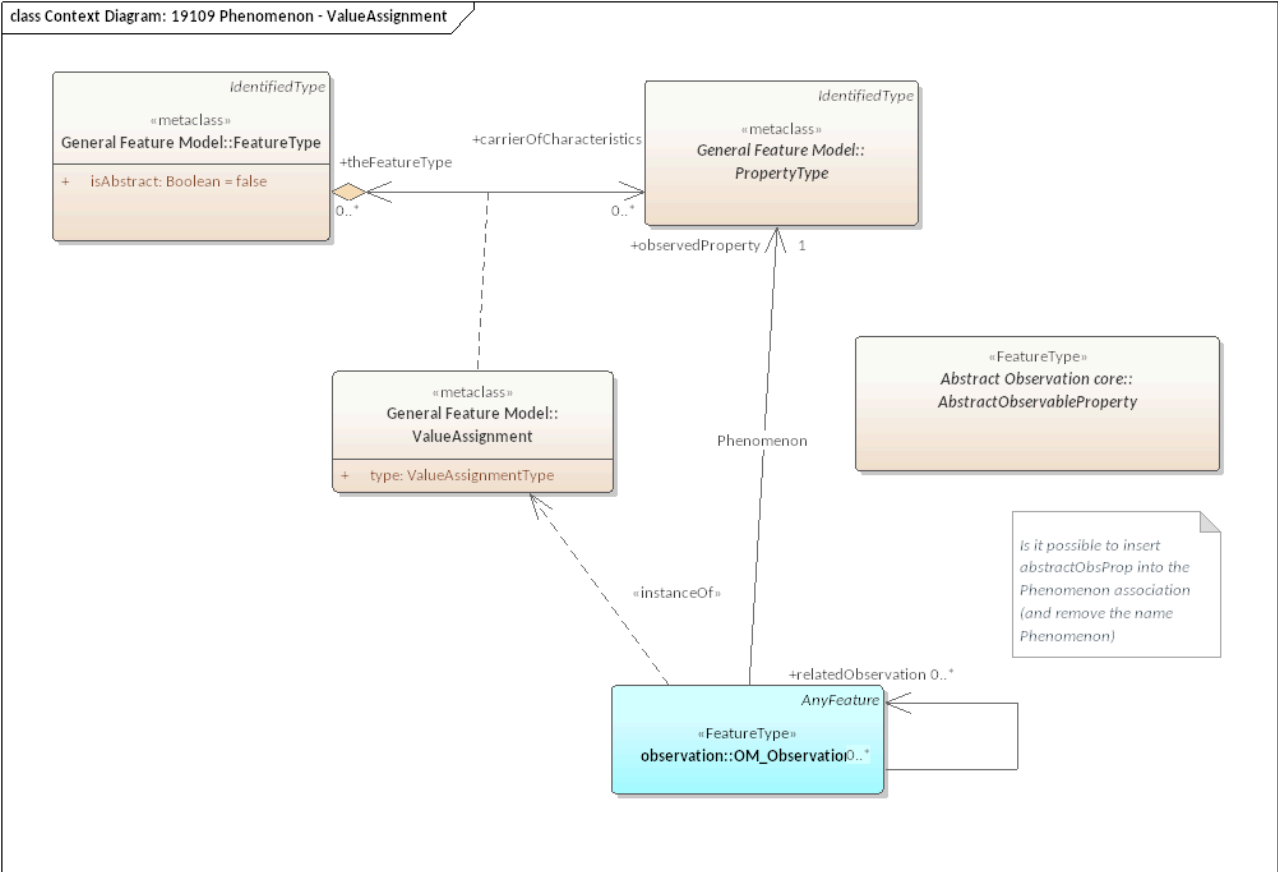


Figure 28 — Context Diagram: 19109 Phenomenon - ValueAssignment

The Examples package is organized into 6 packages:

- a) BasicFeaturesOnly package
- b) Codelist realizations package
- c) Collections package
- d) ModelConsistency package
- e) SamplingCascade package
- f) ModelConsistency — Gender package

11.2 BasicFeaturesOnly package

11.2.1 BasicFeaturesOnly overview

... ..

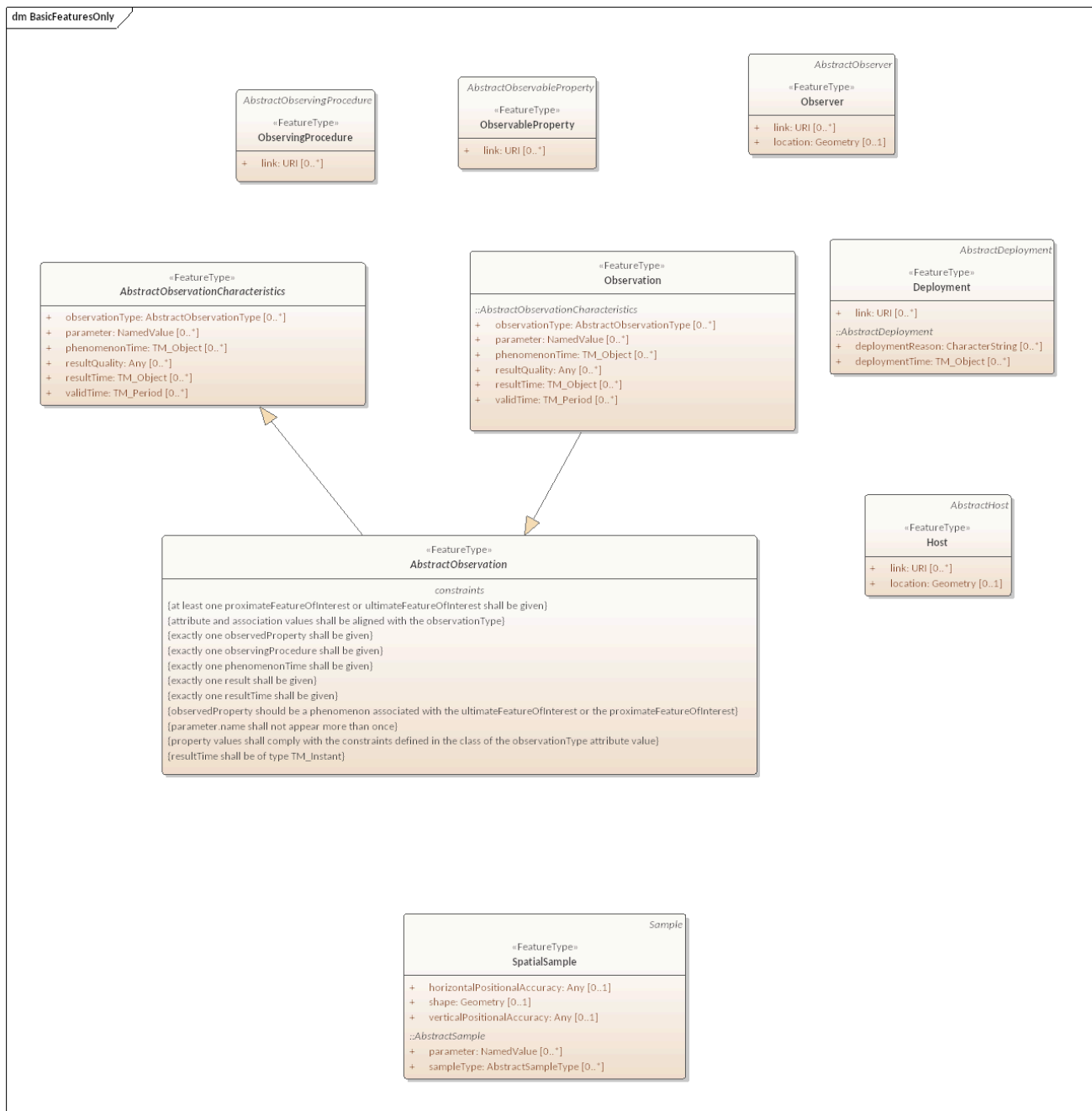


Figure 29 — BasicFeaturesOnly

11.3 Codelist realizations package

11.3.1 Codelist realizations overview

... ..

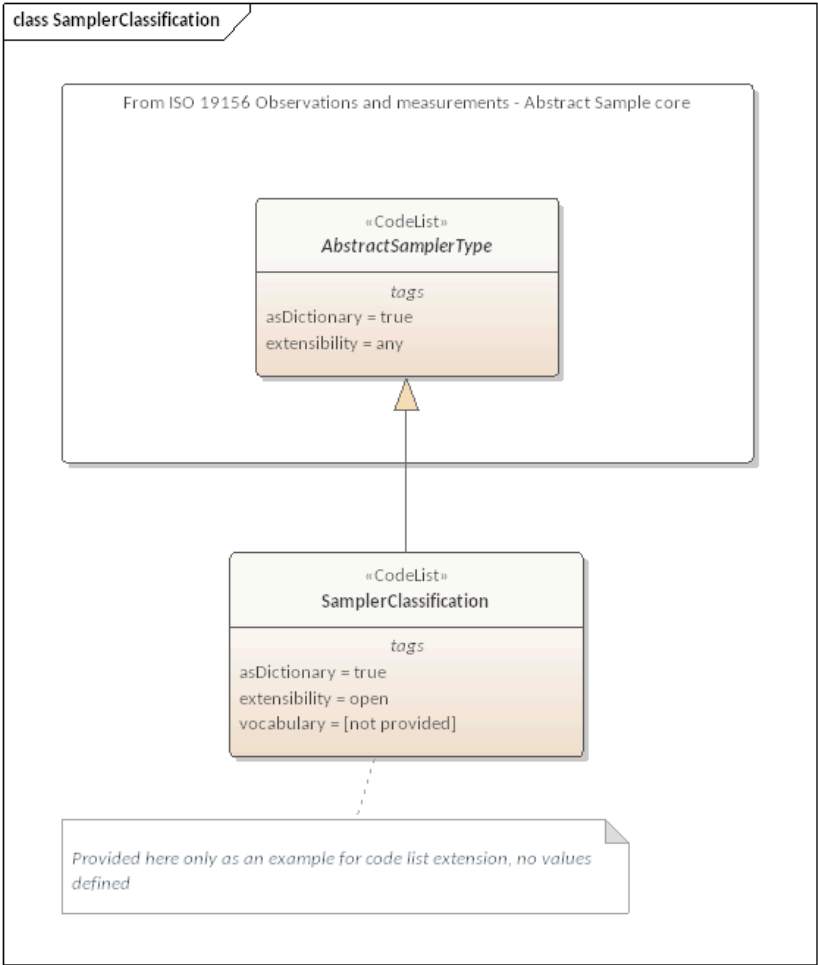


Figure 30 — SamplerClassification

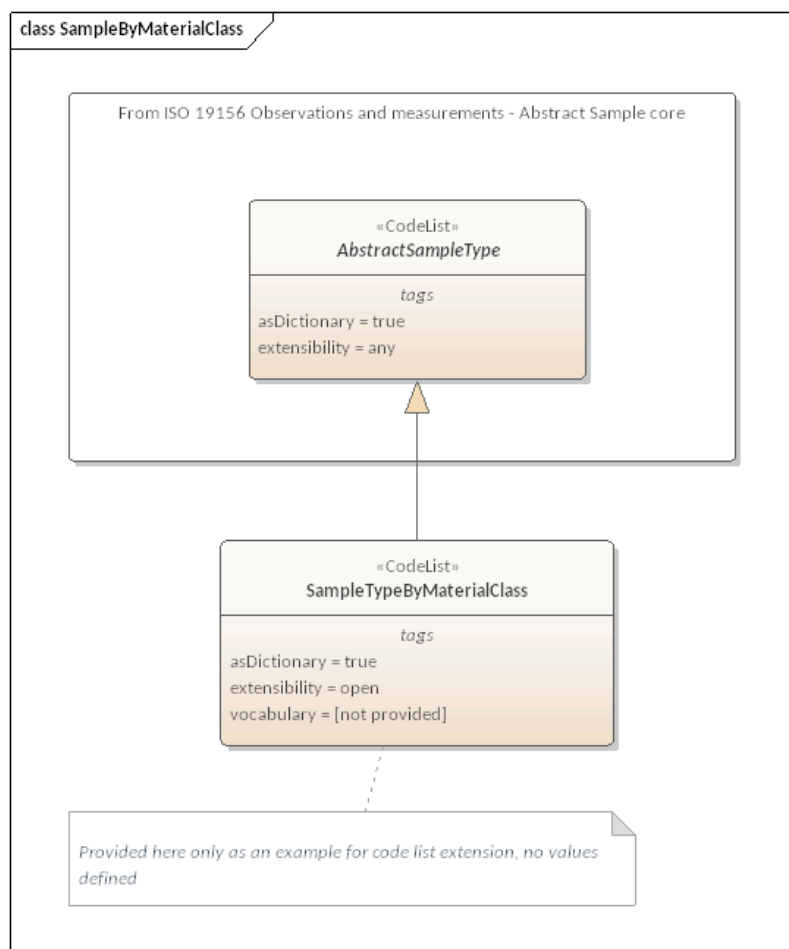


Figure 31 — SampleByMaterialClass

11.3.2 Defining tables

Table 85 — Elements of “Codelist realizations::SamplerClassification” (CodeList)

Name	SamplerClassification
Definition	
Stereotype	CodeList
Inherits from	AbstractSamplerType
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 86 — Elements of “Codelist realizations::SampleTypeByMaterialClass” (CodeList)

Name	SampleTypeByMaterialClass
Definition	AirSample, WaterSample, RockSample, etc.
Stereotype	CodeList
Inherits from	AbstractSampleType
Abstract	False
Associations	(none)
Attributes	(none)

Table 86 (*continued*)

Constraints	(none)
--------------------	--------

**Table 87 — Elements of “Codelist realizations::From ISO 19156
Observations and measurements — Abstract Sample core” ()**

Name	From ISO 19156 Observations and measurements — Abstract Sample core
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

11.4 Collections package

11.4.1 Collections overview

... ..



References

	ObservationCollection
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[illegible]

Table 89 — Elements of “Collections::SampleCollection” ()

Name	SampleCollection
Definition	
Stereotype	interface
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

11.5 ModelConsistency package

11.5.1 ModelConsistency overview

... ..

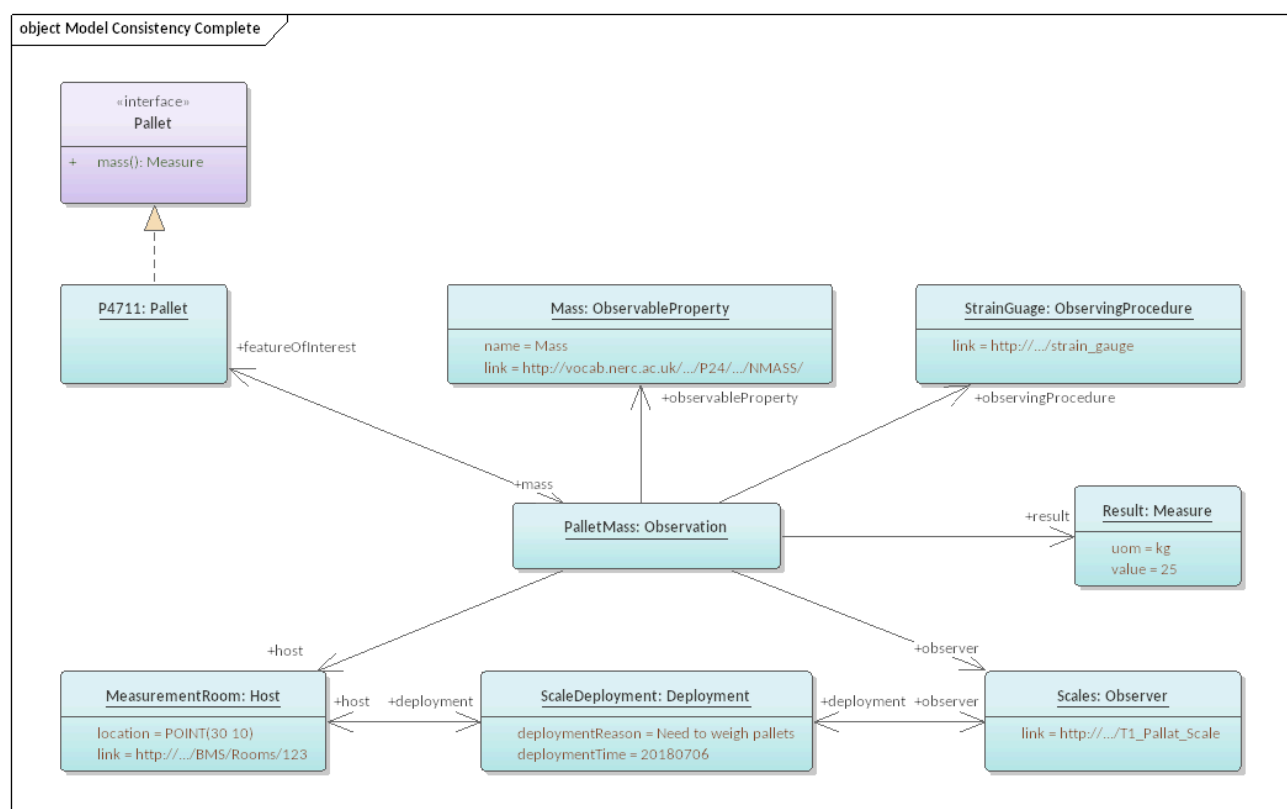


Figure 33 — Model Consistency Complete

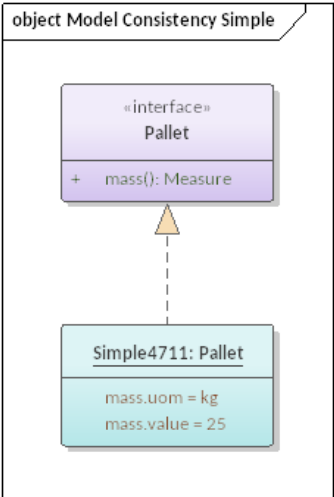


Figure 34 — Model Consistency Simple

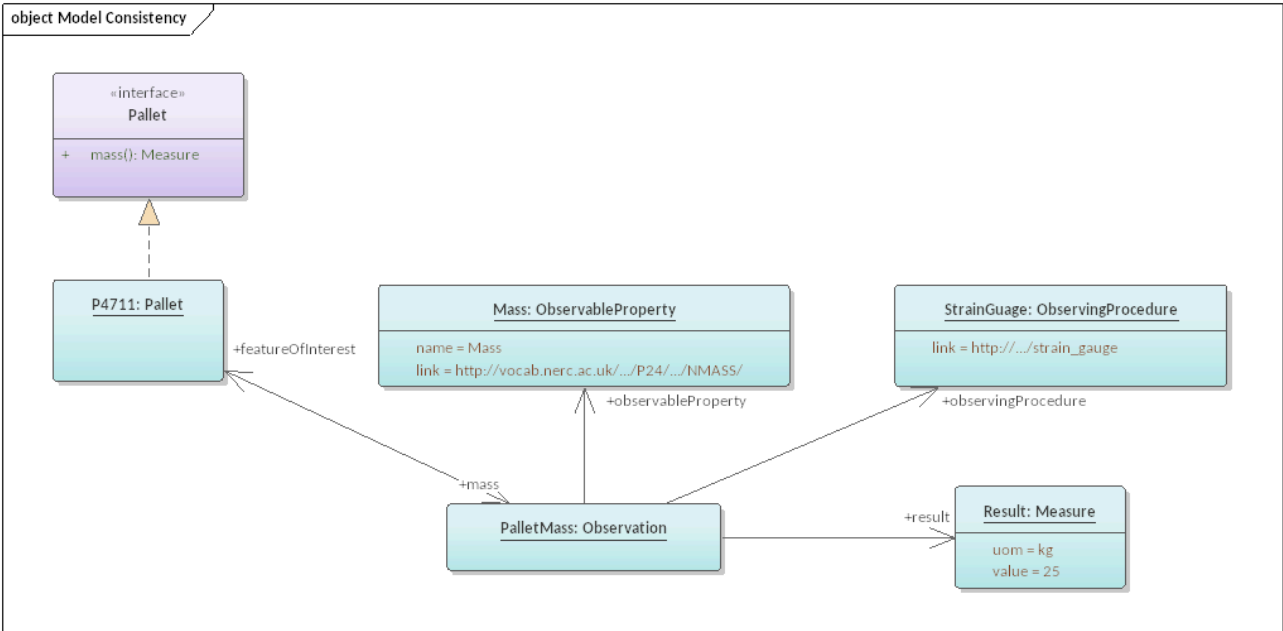


Figure 35 — Model Consistency

11.5.2 Defining tables

Table 90 — Elements of “ModelConsistency::Pallet” ()

Name	Pallet
Definition	
Stereotype	interface
Abstract	True
Associations	(none)
Attributes	(none)
Constraints	(none)

11.6 SamplingCascade package

11.6.1 SamplingCascade overview

... ..

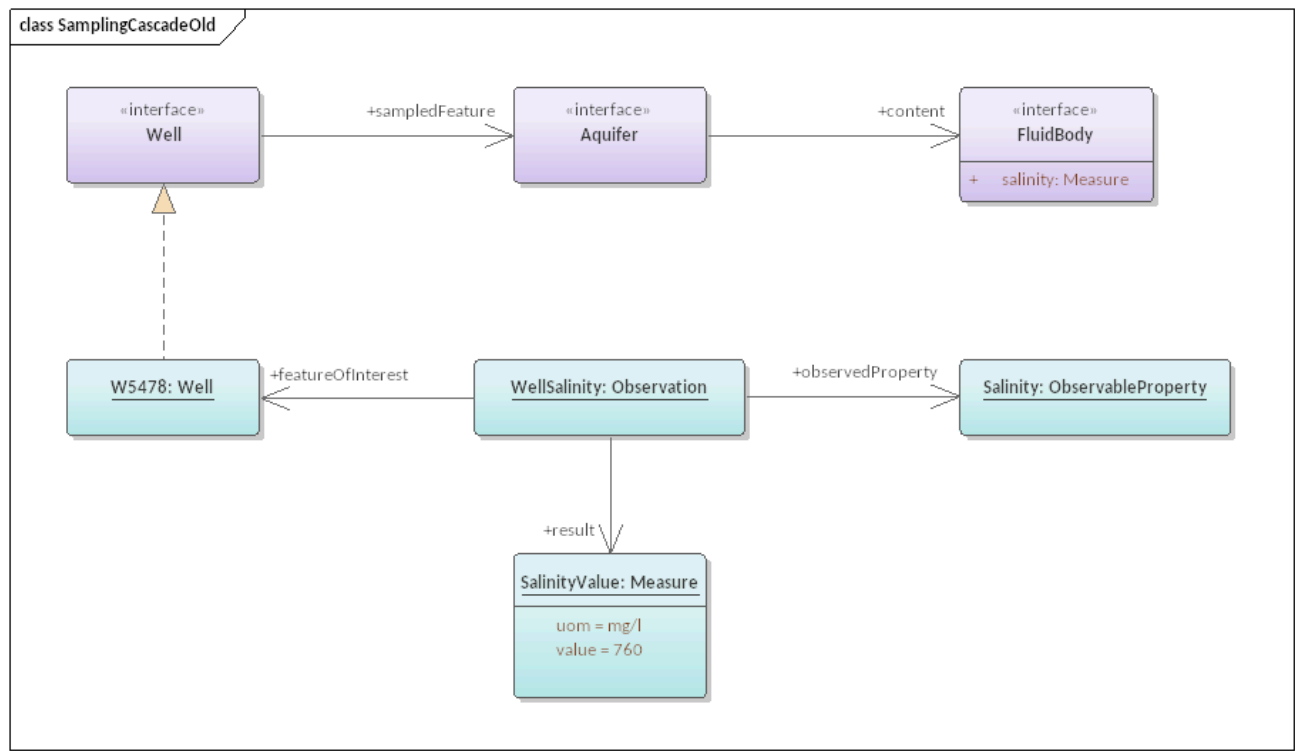


Figure 36 — SamplingCascadeOld

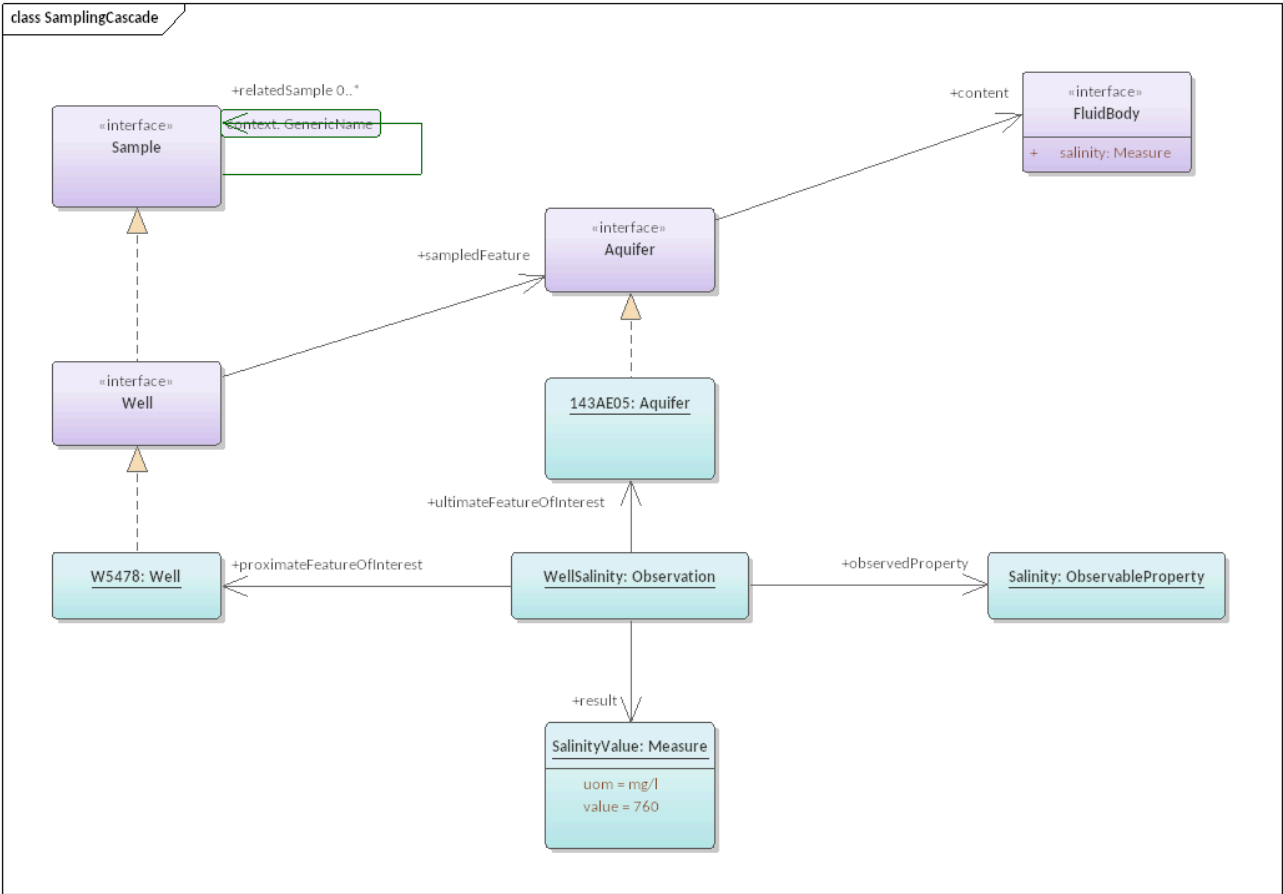


Figure 37 — SamplingCascade

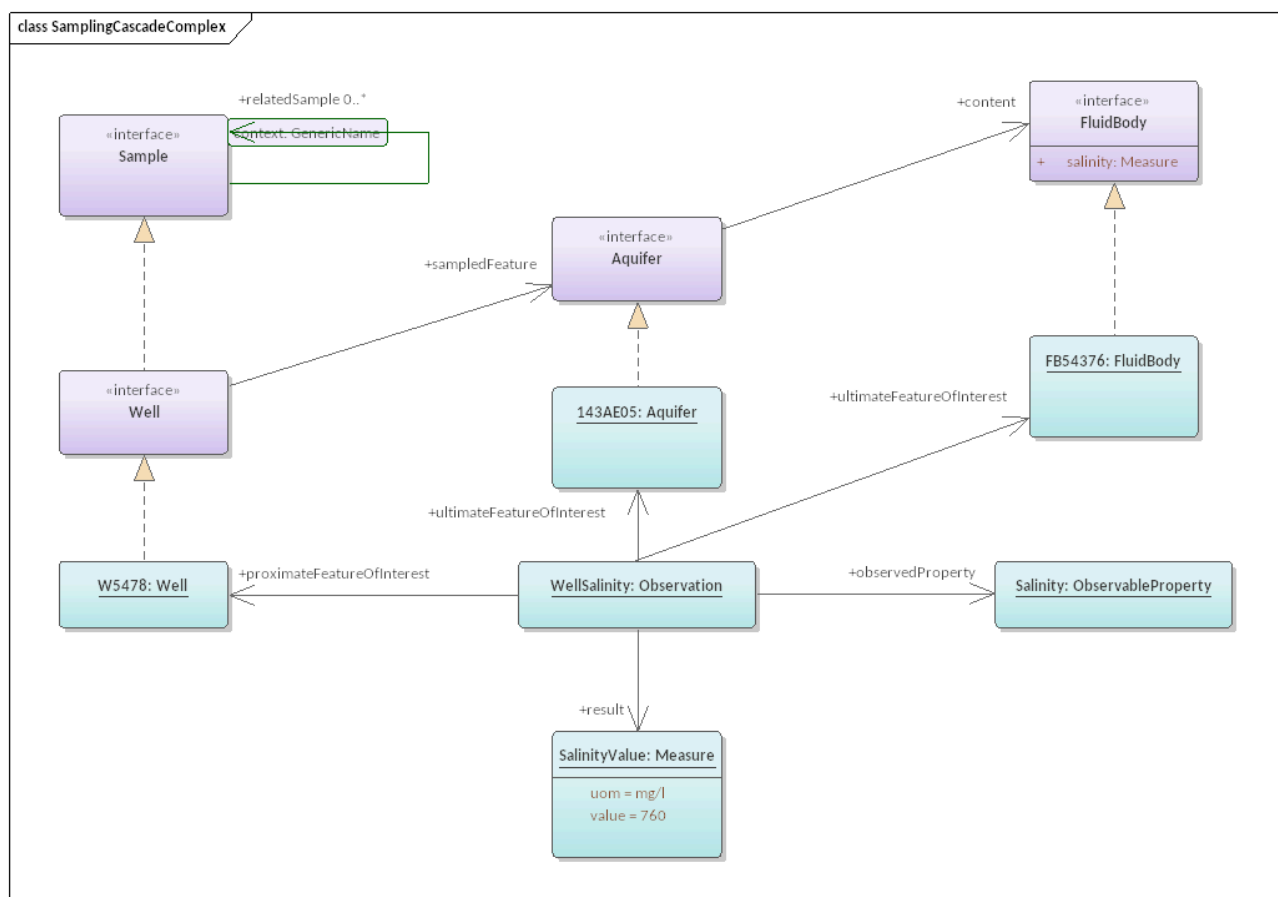


Figure 38 — SamplingCascadeComplex

11.6.2 Defining tables

Table 91 — Elements of “SamplingCascade::W32564” (FeatureType)

Name	W32564
Definition	
Stereotype	FeatureType
Abstract	False
Associations	(none)
Attributes	(none)
Constraints	(none)

Table 92 — Elements of “SamplingCascade::Aquifer” ()

Name	Aquifer		
Definition			
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	FluidBody	FluidBody	(none)
	source:(self) [none]	Aquifer	(none)
	target:content [none]	FluidBody	(none)
	Well	Well	(none)

Table 92 (continued)

	source:(self) [none]	Well	(none)
	target:sampledFeature [none]	Aquifer	(none)
Attributes	(none)		
Constraints	(none)		

Table 93 — Elements of “SamplingCascade::FluidBody” ()

Name	FluidBody		
Definition			
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Aquifer	Aquifer	(none)
	source:(self) [none]	Aquifer	(none)
	target:content [none]	FluidBody	(none)
Attributes	Name	Type	Definition
	salinity [1]	Measure	(none)
Constraints	(none)		

Table 94 — Elements of “SamplingCascade::Well” ()

Name	Well		
Definition			
Stereotype	interface		
Abstract	True		
Associations	Name	Type	Definition
	Aquifer	Aquifer	(none)
	source:(self) [none]	Well	(none)
	target:sampledFeature [none]	Aquifer	(none)
Attributes	(none)		
Constraints	(none)		

11.7 ModelConsistency — Gender package

11.7.1 ModelConsistency — Gender overview

... ..

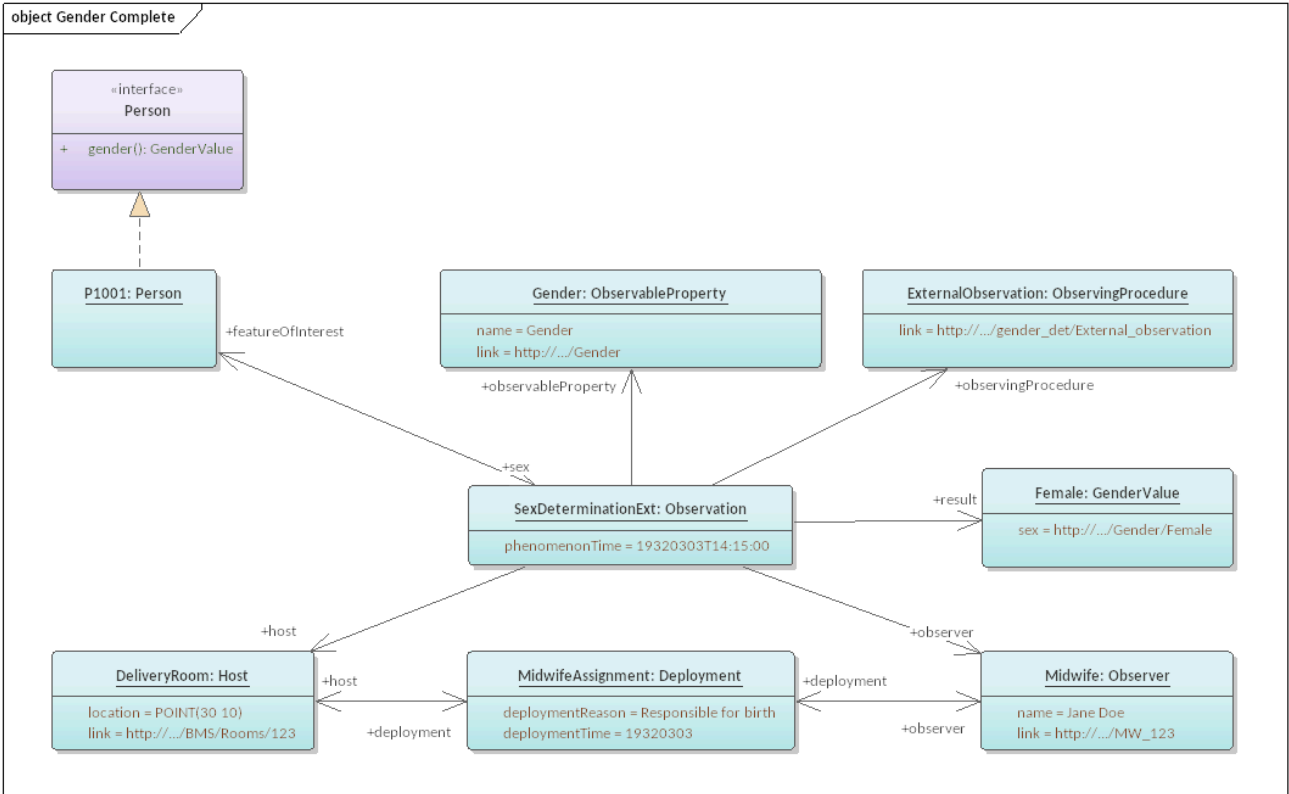


Figure 39 — Gender Complete

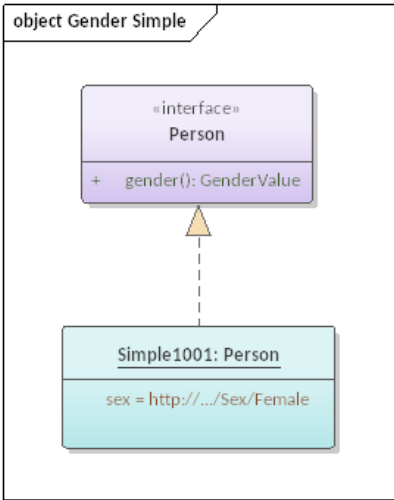


Figure 40 — Gender Simple

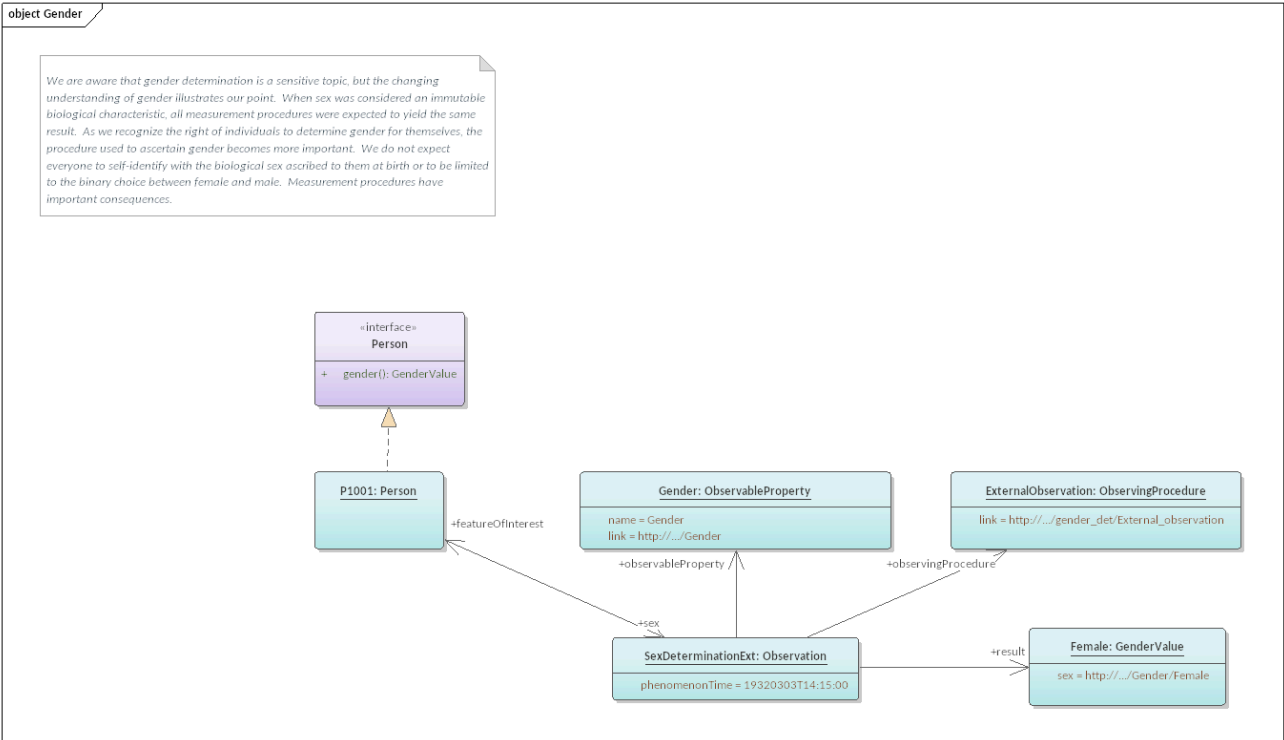


Figure 41 — Gender

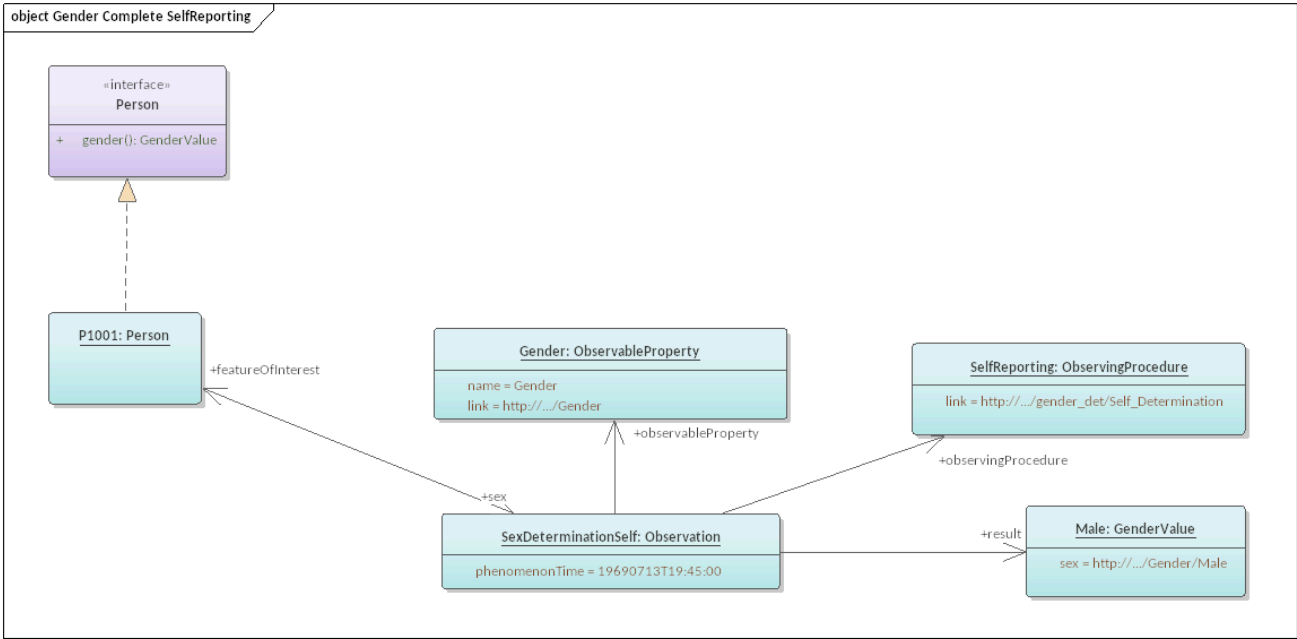


Figure 42 — Gender Complete SelfReporting

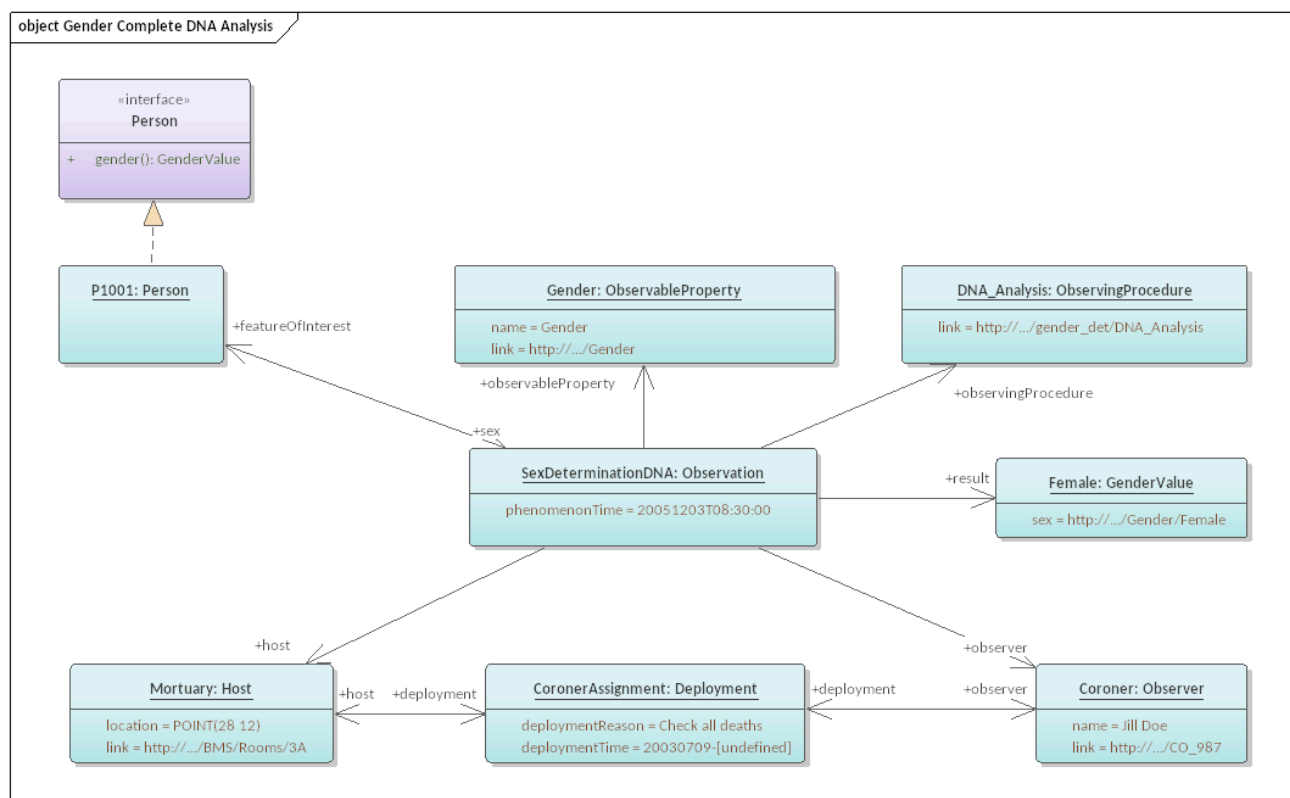


Figure 43 — Gender Complete DNA Analysis

11.7.2 Defining tables

Table 95 — Elements of “ModelConsistency — Gender::GenderValue” ()

Name	GenderValue		
Definition			
Stereotype	interface		
Abstract	False		
Associations	(none)		
Attributes	Name	Type	Definition
	sex [1]	String	(none)
Constraints	(none)		

Table 96 — Elements of “ModelConsistency — Gender::Person” ()

Name	Person
Definition	
Stereotype	interface
Abstract	True
Associations	(none)
Attributes	(none)
Constraints	(none)

Bibliography

